

A Primer on Dependency in Polynomial Product: Identify, Exploit, and Trim

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Abstract. Many lattice-based encryption schemes admit a negligible but nonzero decryption failure rate (DFR), which is tied to both correctness and security through failure-based attacks. Several module-lattice constructions (e.g., LAC at NIST PQC Round 2 and DAWN at Asiacrypt 2025), as well as average-case noise analyses in FHE, estimate the DFR from one-coordinate marginals combined with an independence approximation across the coefficients of polynomial products. Geometrically, this approximation models the noise as uniformly distributed on a sphere. In the rare-event regime relevant to concrete security, the approximation can be optimistic: polynomial convolution introduces structured dependencies with no analogue in unstructured lattice settings, and the true noise spreads towards a cube rather than a sphere.

To make this effect explicit, we study polynomial products in power-of-two cyclotomic rings through a norm-wise decomposition. The decomposition separates an *outer* term (corresponding to the sphere’s radius), which is effectively captured by coefficient-wise models, from an *inner* term (representing the uneven parts of the spherical surface) that measures convolution-induced dependencies. This gives an exact account of the heavier tails observed in polynomial products and of the gap between independence-based estimates and actual failure behavior.

This perspective has consequences for both attacks and design. On the attack side, it yields a principled proxy criterion for constructing high-DFR candidate ciphertexts in failure-based attacks. In particular, it explains why the attack of Guo *et al.* (Asiacrypt 2019) remains effective against LAC under fixed Hamming-weight sampling, and it improves failure-finding efficiency by identifying the underlying class of bad randomness pairs beyond pattern-based subsets. On the design side, it motivates trimming high-dependency samples during key generation and encryption. We provide a certified trimmed DFR bound based on conditional spectral control and a complementary three-vector heuristic DFR bound with experimental validation for calibrated interpretation. We formalize the resulting approach as generic frameworks **TrimPKE** and **TrimKEM**, prove security in the QROM while accounting for rejection, and instantiate them for LAC and DAWN as case studies.

Keywords: module lattice · decryption failures · PKE · KEM

1 Introduction

As post-quantum cryptography moves toward standardization, lattice-based constructions have gained increasing prominence due to their efficiency, relatively direct reductions to well-studied hardness assumptions, and applicability across a wide range of cryptographic primitives. In particular, the lattice-based KEM Kyber [6] has been selected for standardization by NIST. A distinctive feature of many lattice-based encryption schemes, however, is that decryption may fail with small probability. This phenomenon is common to constructions based on Learning with Errors (LWE) and its ring/module variants. While one can eliminate decryption failures through sufficiently conservative parameter choices as in NTRU Prime [4] and NTRU HRSS [34], this typically incurs noticeable bandwidth costs (e.g., larger public keys and ciphertexts). Consequently, modern KEM designs tune parameters so that the decryption failure rate (DFR) is negligible while retaining performance, as exemplified by Kyber [6] and DAWN [42]. In fully homomorphic encryption (FHE), DFR also plays an analogous role: schemes such as BGV [9], BFV [8,22], and CKKS [12,7] estimate and control failure probabilities to ensure functional correctness of the supported evaluations.

DFR also has direct security implications in the chosen-ciphertext (CCA) model. An adversary with access to a decryption oracle can exploit failures to obtain information about the secret key, making careful parameterization essential for balancing efficiency and security. For example, the NIST PQC setting allows up to 2^{64} adversarially chosen ciphertexts, and security against quantum adversaries typically calls for DFR well below 2^{-128} . Failure-based CCA attacks on code/lattice-based schemes have been extensively studied in [21,28,29,33]. Moreover, the advent of failure boosting [19] and subsequent refinements [20] reduces the dominant attack cost to finding the first failing ciphertext, further increasing the importance of accurate tail estimates. Since the introduction of IND-CPA-D security [39], DFR has come to play a more important role in the security of FHE schemes; several failure-based attacks on FHE schemes have been studied in [10,11,44].

Error-correcting codes (ECCs) are a standard tool for reducing ciphertext and key sizes in module-lattice-based encryption; LAC [45] (NIST PQC Round 2) is a representative example. Underestimating the DFR in such schemes, however, leaves them vulnerable to failure-based CCA attacks [29]. In a related direction, average-case noise analysis is common for estimating DFR in FHE schemes, as in [5]. The DFR estimation used in LAC [45] and DAWN [42], as well as the average-case noise estimation common in FHE [5,14], share a common assumption: the relevant post-convolution coefficients are treated as independent. Several works provide empirical evidence consistent with this assumption in typical regimes [14,41], but these evaluations largely focus on matching the coarse shape of the probability mass function (PMF), rather than quantifying the statistical distance to the true distribution in the extreme tail. For CKKS, for example, the analysis of [7] reports that the mean and standard deviation of the noise deviate from theoretical predictions by at most 0.01 bits; such moment-level agreement, however, does not rule out substantial discrepancies in rare-

event probabilities. Indeed, [25] observed that independence-based models can induce heavy-tailedness in certain settings and analyzed higher moments (e.g., kurtosis), revealing a non-negligible bias relative to independence-based estimates. Further, [15] demonstrated that convolution-induced dependencies can lead to underestimation of DFR in schemes that combine module-lattice noise with error-correcting codes, such as LAC [45]. That work quantified a gap between independence-based estimates and dependence-aware estimates for LAC. That work, however, still relied on additional independence-style assumptions and did not capture dependency effects under fixed Hamming-weight samplers, precisely the regime in which LAC operates.

1.1 Our Contributions

Several DFR analyses for module-lattice encryption rely on a coefficient-wise independence assumption for polynomial products, which can lead to systematic underestimation of the DFR. We study the dependencies induced by polynomial product and develop a norm-wise framework that both exploits and mitigates their impact. Our goal is to identify where these dependencies matter, exploit them to shape a better failure-based attack, and quantify how much trimming is required to suppress the resulting tail amplification in ECC-relevant settings. Our contributions are as follows.

Identify: a norm-wise dependency model. We introduce a norm-wise perspective that makes the dependency contribution explicit and distinguishes module lattices from the unstructured lattice setting in the tail regime. Using a new decomposition-based view, we separate the behavior explained by coefficient-wise models from the additional effect induced by polynomial convolution. This clarifies when and why independence-based estimators become inaccurate for DFR in schemes where failures are driven by many moderately large coefficients.

Exploit: a dependency-guided failure-finding heuristic. On the attack side, our model explains why failure-based CCA attacks can still find useful candidates even when the relevant Hamming weights are fixed, and it recovers prior pattern-based selection rules as a particularly structured example. It suggests a dependency-guided proxy selector for high-risk candidates. In our small-scale proxy experiment, this selector exhibits a stronger tail signal than the prior pattern proxy under the same search budget.

Trim: a generic protection framework. On the design side, we propose a lightweight mitigation based on rejection sampling that filters highly dependent samples during key generation and encryption. We present a trimming-conditioned worst-case DFR bound with spectral control, while the sharper three-vector estimate is kept as a separate heuristic supplement. We formalize this approach as two generic frameworks, TrimPKE and TrimKEM, prove their

security in the QROM while accounting for rejection, and explicitly analyze the resulting concrete-security loss. We use the LAC and DAWN frameworks as case studies to illustrate the trade-offs between DFRs and efficiency.

1.2 Our Techniques

Lattice-based encryption schemes differ in the key-generation assumption they rely on (NTRU- or LWE-type), but their encryption and decryption procedures are uniformly governed by LWE or one of its structured variants (Ring/Module-LWE, and the corresponding LWR variants). This common structure admits a generic decryption condition. Concretely, decryption recovers each message bit m_i from a noisy value of the form $\langle \mathbf{a}_i, \mathbf{b}_i \rangle + \frac{q}{2}m_i$, where $\mathbf{a}_i, \mathbf{b}_i$ are (public/secret-derived) vectors and $\langle \mathbf{a}_i, \mathbf{b}_i \rangle$ is the effective noise term. Let $z \in \mathcal{R}_n$ denote the centered pre-decoding error polynomial, let $\Delta > 0$ denote the per-coordinate decision threshold, and let

$$N_\Delta := \#\{i \in \{0, \dots, n-1\} : |z_{[i]}| > \Delta\}$$

count the threshold exceedances. For schemes without ECC, the failure event is $\|z\|_\infty > \Delta$, which equals $N_\Delta = 0$; with a hard-decision code correcting up to t coefficient errors, the failure event becomes $N_\Delta > t$. Accordingly, bounding the DFR reduces to understanding the joint tail behavior of these inner products rather than only their one-coordinate marginal law.

To obtain compact keys and ciphertexts and to enable fast arithmetic, many modern designs (e.g., Kyber [6], Saber [16], BAT [24], NEV [51], and DAWN [42]) adopt Ring-/Module-LWE (or Ring-/Module-LWR). In these structured settings, vector entries are coefficient embeddings of ring elements. For a fixed component index k , the entries $(\mathbf{a}_i)_{[k]}$ and $(\mathbf{a}_j)_{[k]}$ are not independent draws: they are rotations of the same underlying polynomial in a quotient ring $\mathbb{Z}[x]/\varphi$, satisfying $(\mathbf{a}_i)_{[k]} \equiv x^{j-i} \cdot (\mathbf{a}_j)_{[k]}$. As a consequence, the family of inner products appearing across coordinates in decryption corresponds to different coefficients of the *same* polynomial products. This is the algebraic source of the dependencies that are absent in unstructured lattices, yet are often ignored by analyses that model coefficients as independent.

Making the dependency contribution explicit. To isolate this effect, we view the centered decryption noise schematically as

$$z = \sum_i a_i b_i + \sum_j w_j,$$

where a_i, b_i, w_j are ring elements sampled independently according to the scheme. The dependence of interest arises from the products $a_i b_i$: their coefficient vectors are *not* well approximated by independent coordinates. We capture this phenomenon through a norm-wise comparison between polynomial products and their unstructured matrix analogues. Since most practical module-lattice

schemes use power-of-two cyclotomic rings, we focus on $\mathbb{Z}[x]/(x^n + 1)$. For $a, b \in \mathbb{Z}[x]/(x^n + 1)$ and $\bar{a}(x) = a(x^{-1})$, we use the following exact identity:

$$\|ab\|_2^2 = \boxed{\|a\|_2^2 \|b\|_2^2} + \boxed{2\langle (a\bar{a})_{[1:n/2]}, (b\bar{b})_{[1:n/2]} \rangle},$$

where $(a\bar{a})_{[1:n/2]}$ denotes the vector of coefficients of $a\bar{a}$ from degrees 1 through $n/2 - 1$. The conceptual point is the following: coefficient-wise, independence-based analyses effectively capture the first (solid-box) term, while the second (dashed-box) term, the convolution-induced dependency carrier, is exactly what they miss. Even when the coefficients of a and b have mean zero so that the dashed-box term has expectation 0, it can contribute substantial tail mass to $\|ab\|_2^2$, producing occasional large spikes that are invisible to moment-level checks. This explains the heavier tails observed empirically in polynomial products and accounts for the underestimation of rare decryption failures incurred by independence-based analyses (e.g., module-lattice encryption with ECC and average-case FHE noise analyses). Furthermore, based on such a dependency-aware model, we provide a rigorous upper bound of DFR via canonical embedding and a heuristic estimate of DFR under a three-vector model with experimental validation in Section 3.3.

Improved failure-finding heuristic. The norm-wise perspective yields a concrete notion of “bad” randomness pairs for schemes whose decoding failures are driven by multiple coefficients above threshold. In particular, the previous analysis in [15] tracks only the outer behavior (captured by the solid-box term), which may fail to explain attacks that remain effective under fixed Hamming-weight sampling, as in the LAC setting [29]. Under the identity above, the explanation is straightforward: even when the outer contribution is essentially fixed, the dependency carrier can fluctuate significantly, thereby increasing the relevant tail risk. This viewpoint also clarifies why the patterns exploited in [29] are effective: they correlate with unusually large values of $(a\bar{a})_{[1:n/2]}$ (and thus a large dependency contribution). Consequently, instead of restricting to a specific pattern family, we can target the underlying cause and obtain a broader selection rule for high-risk candidates. Section 4 evaluates this claim with explicit tail statistics.

Mitigation via trimming. Finally, we use the same model to design a countermeasure that reduces dependency-induced tail amplification without changing the original parameter sets. The key observation is that the dependency carrier is reflected in simple statistics of $\|(a\bar{a})_{[1:n/2]}\|_\infty$; thus, we can filter atypical high-dependency samples by rejection sampling. Section 3.3 first turns this idea into a certified worst-case DFR bound for accepted samples and then gives a separate heuristic supplement for finer-grained interpretation and experimental backup. This rejection sampling mechanism is lightweight, preserves the original scheme structure, and admits a clean security analysis: the only effect is an additional rejection probability that must be accounted for in reductions and concrete bounds. In Section 5, we formalize trimming as generic frameworks (TrimPKE

and TrimKEM), prove their security in the QROM while explicitly incorporating rejection, and evaluate instantiations for LAC and DAWN, demonstrating practical efficiency-robustness trade-offs in Table 1.

2 Preliminaries

2.1 Notations

We use $\mathcal{R}_n$ to denote the cyclotomic ring $\mathbb{Z}[x]/(x^n + 1)$, where n is a power of 2, and we write $\mathcal{R}_{n,q} := \mathcal{R}_n/q\mathcal{R}_n$. We use $\mathbb{Z}_q$ to represent the centered residue set $\{-\frac{q-1}{2}, \dots, 0, \dots, \frac{q-1}{2}\}$, where q is an odd prime. We denote by regular font letters (e.g. a, b) the elements of $\mathcal{R}_n$, by bold lower-case letters (e.g. $\mathbf{a}, \mathbf{b}$) vectors over $\mathbb{Z}$ or $\mathcal{R}_n$, and by bold upper-case letters (e.g. $\mathbf{A}, \mathbf{B}$) matrices whose entries lie in $\mathcal{R}_n$. For a polynomial a , we denote the coefficient of x^i by $a_{[i]}$, and we use $\bar{a}$ to represent the conjugate polynomial $\bar{a}(x) = a(x^{-1})$. Additionally, we denote the vector formed by the coefficients $(a_{[i]}, a_{[i+1]}, \dots, a_{[j-1]})$ as $a_{[i:j]} \in \mathbb{Z}^{j-i}$. We write $\|a\|_2 := \|a_{[0:n]}\|_2$. We refer to the squared ℓ_2 -norm $\|a\|_2^2$ as the *outer distortion*, and to $\|(a\bar{a})_{[1:n/2]}\|_\infty$ as the *inner distortion*. For $r \in \{0, \dots, n-1\}$, let $\omega_r := \exp(\frac{(2r+1)\pi i}{n})$ denote the negacyclic evaluation points. We use the function $\text{Sample}(\mathcal{X})$ to denote sampling from the distribution $\mathcal{X}$. We use the notation $\mathcal{X}^n$ for the distribution of polynomials in $\mathcal{R}_n$ whose coefficients are drawn from $\mathcal{X}$, and $\mathcal{U}(\mathcal{V})$ for the uniform distribution on the set $\mathcal{V}$.

2.2 Lattice-based Encryption

For simplicity, we adopt a generic encryption framework based on Module-LWE (e.g., Kyber [6]) as a case study, focusing on the high-level structure rather than implementation details, as shown in Figure 1. Similarly, we consider a generic NTRU-based framework (e.g., NTRU-HPS [30]) at a comparable level of abstraction in Figure 2.

2.3 Hard-Decision Error-Correcting Codes

Let $z \in \mathcal{R}_n$ denote the centered decryption-error polynomial before the final decision rule is applied, and let $\Delta > 0$ be the per-coordinate decoding threshold. A coordinate error occurs at index i when $|z_{[i]}| > \Delta$. For schemes without an outer ECC layer, decryption therefore fails exactly when

$$\|z\|_\infty > \Delta.$$

For schemes with a hard-decision code that corrects up to t coefficient errors, we define

$$N_\Delta := \#\{i \in \{0, \dots, n-1\} : |z_{[i]}| > \Delta\},$$

so the ECC failure event is $\{N_\Delta > t\}$.

Fig. 1: **Module-LWE-based Encryption**

KeyGen	Encrypt($pk = (\mathbf{A}, \mathbf{b}), m$)
1 : $\mathbf{A} \leftarrow \text{Sample}(\mathcal{U}(\mathcal{R}_{n,q})^{l \times k})$	1 : $\mathbf{r} \leftarrow \text{Sample}((\mathcal{X}_r^n)^l)$
2 : $\mathbf{s} \leftarrow \text{Sample}((\mathcal{X}_s^n)^k)$	2 : $\mathbf{e}_1 \leftarrow \text{Sample}((\mathcal{X}_{e_1}^n)^k)$
3 : $\mathbf{e} \leftarrow \text{Sample}((\mathcal{X}_e^n)^l)$	3 : $\mathbf{e}_2 \leftarrow \text{Sample}(\mathcal{X}_{e_2}^n)$
4 : $\mathbf{b} \leftarrow \mathbf{s}\mathbf{A}^\top + \mathbf{e} \pmod{q}$	4 : $\mathbf{c}_1 \leftarrow \mathbf{r}\mathbf{A} + \mathbf{e}_1 \pmod{q}$
5 : $pk \leftarrow (\mathbf{A}, \mathbf{b})$	5 : $c_2 \leftarrow \mathbf{r}\mathbf{b}^\top + e_2 + \lfloor \frac{q}{2} \rfloor \cdot m \pmod{q}$
6 : $sk \leftarrow \mathbf{s}$	6 : return $(\mathbf{c}_1, c_2)$
7 : return (pk, sk)	
Decrypt($\mathbf{c}_1, c_2, sk = \mathbf{s}$)	
1 : $m \leftarrow \lfloor \frac{2}{q} \cdot (c_2 - \mathbf{c}_1 \mathbf{s}^\top \pmod{q}) \rfloor \pmod{2}$	
2 : return m	

Fig. 2: **NTRU-based Encryption**

KeyGen	Encrypt($pk = h, m$)
1 : $f' \leftarrow \text{Sample}(\mathcal{X}_f^n)$	1 : $t \leftarrow \text{Sample}(\mathcal{X}_t^n)$
2 : $g \leftarrow \text{Sample}(\mathcal{X}_g^n)$	2 : $c \leftarrow p \cdot ht + m \pmod{q}$
3 : $f \leftarrow p \cdot f' + 1$	3 : return c
4 : if f is not invertible on $\mathcal{R}_n/q\mathcal{R}_n$ go back to step 1	
5 : $h \leftarrow gf^{-1} \pmod{q}$	
6 : $pk \leftarrow h$	
7 : $sk \leftarrow f$	
8 : return (pk, sk)	
Decrypt($c, sk = f$)	
	1 : $c' \leftarrow cf \pmod{q}$
	2 : $m \leftarrow c' \pmod{p}$
	3 : return m

In the Module-LWE framework of Figure 1, the centered decryption error is

$$z := \mathbf{r}\mathbf{e}^\top - \mathbf{e}_1\mathbf{s}^\top + e_2,$$

and the standard rounding decoder corresponds to $\Delta = \lfloor q/4 \rfloor$ after centering. The same notation will be used throughout the paper: z is always the centered pre-decoding error polynomial, Δ is always the per-coordinate threshold, and N_Δ is always the number of threshold exceedances.

3 Identify Dependencies in Polynomial Convolution

"The closer you look, the less you see."

— *Now You See Me*

In this section, we review existing methods for estimating decryption failure rates (DFR) in module-lattice-based encryption schemes that rely on the independence assumption and highlight their limitation. We then develop the exact norm-wise identities that isolate the relevant dependency terms, explain why inner-distortion statistics are natural trimming proxies, and use centered rejection-sampling experiments to illustrate how trimming suppresses the tail amplification. Finally, we provide the analysis that bridges the quantification of dependency to a concrete DFR estimate.

3.1 Inaccuracy of Independence Assumption

We first review the standard independence-based DFR estimate and pinpoint where it ceases to describe the true joint law. We use the generic Module-LWE framework of Figure 1, with ring $\mathcal{R}_n$. During decryption, one computes

$$c_2 - \mathbf{c}_1 \mathbf{s}^\top \equiv \mathbf{r} \mathbf{e}^\top - \mathbf{e}_1 \mathbf{s}^\top + e_2 + \left\lfloor \frac{q}{2} \right\rfloor \cdot m \pmod{q}. \quad (1)$$

Let

$$z \triangleq \mathbf{r} \mathbf{e}^\top - \mathbf{e}_1 \mathbf{s}^\top + e_2$$

be the resulting centered error polynomial. Without an outer ECC layer, the failure event is $\|z\|_\infty > \Delta$ with $\Delta = \lfloor q/4 \rfloor$; with a t -error-correcting hard-decision code, the failure event is $N_\Delta > t$ as defined in Section 2.3.

The standard estimation method begins by computing the marginal distribution $\mathcal{X}_z$ of a single coefficient of z . The distribution of e_2 is straightforward, as its coefficients are independent by design. In contrast, the convolution terms $\mathbf{r} \mathbf{e}^\top$ and $\mathbf{e}_1 \mathbf{s}^\top$ are more involved. In the no-ECC setting, the conventional estimate is the union bound

$$\text{DFR} = \Pr[\|z\|_\infty > \Delta] \leq \sum_{i=0}^{n-1} \Pr[|z_{[i]}| > \Delta] = n \cdot \Pr_{x \leftarrow \mathcal{X}_z}[|x| > \Delta]. \quad (2)$$

This bound matches the no-ECC decision rule but does not describe schemes with ECC, where several moderately large coefficients can jointly trigger failure. The standard workaround is to replace the joint law of the indicators N_Δ by an i.i.d. Bernoulli model with parameter $p_\Delta := \Pr_{x \leftarrow \mathcal{X}_z}[|x| > \Delta]$. For example, if the ECC corrects up to t coefficient errors, the resulting binomial model becomes

$$\text{DFR} = 1 - \sum_{i=0}^t \binom{n}{i} \left(\Pr_{x \leftarrow \mathcal{X}_z}[|x| > \Delta] \right)^i \left(1 - \Pr_{x \leftarrow \mathcal{X}_z}[|x| > \Delta] \right)^{n-i}, \quad (3)$$

where t is the maximum number of errors the ECC used can correct. This expression is an independence-based approximation, not a theorem about the true

DFR. In practice, the marginal law $\mathcal{X}_z$ is often computed numerically (e.g., via scripts used in Kyber [6]). The main limitation is therefore not the one-coordinate marginal, but the replacement of the joint law by independent Bernoulli trials. While the central limit theorem provides asymptotic guidance under independence as $n \rightarrow \infty$, the target DFRs are typically exponentially small (e.g., below 2^{-128}). In this regime, even weak dependence can materially change the tail probability of N_Δ .

To illustrate the effect of structural dependencies, let $a, b \in \mathcal{R}_n$. Although the pairwise covariance of distinct coefficients in the product ab can vanish on average for centered samplers [41], this does not imply that the joint tail behaves as if the coefficients were independent. We therefore study the squared ℓ_2 -norm as a structural diagnostic: it aggregates many moderately large coefficients and reacts to the same kind of coefficient clustering that later enters the ECC event $N_\Delta > t$, even though it is not itself the failure criterion. Specifically, we sample 2^{32} polynomial pairs $(a, b) \in \mathcal{R}_{256} \times \mathcal{R}_{256}$ with coefficients drawn uniformly from $\mathbb{Z}_3$, and compare the squared ℓ_2 -norm of their products against 2^{32} matrix-vector products $\mathbf{A}\mathbf{b}$ with $(\mathbf{A}, \mathbf{b}) \in \mathbb{Z}^{256 \times 256} \times \mathbb{Z}^{256}$ whose entries are independent and uniform on $\mathbb{Z}_3$. The comparison is performed over integer products before reduction modulo q , so that the plot isolates convolution-induced dependence rather than wrap-around effects.

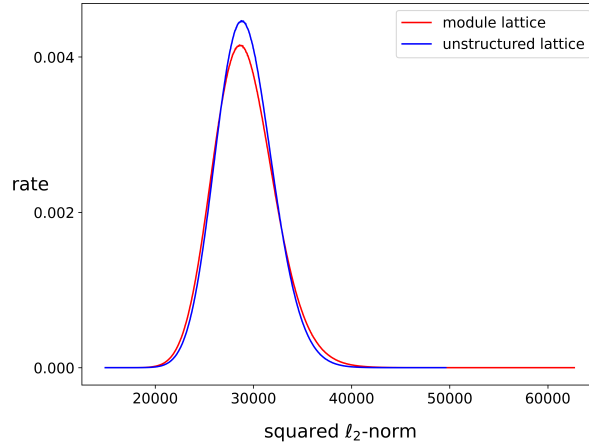


Fig. 3: Empirical distributions of $\|ab\|_2^2$ for integer negacyclic products (red) and $\|\mathbf{A}\mathbf{b}\|_2^2$ for an i.i.d. matrix baseline (blue).

Figure 3 highlights a clear discrepancy: polynomial products inherit hidden coefficient-wise dependencies corresponding to the module lattice, while the matrix case represents a genuine i.i.d. baseline corresponding to the unstructured lattice. The x-axis shows the squared ℓ_2 -norm, and the y-axis shows the frequency of observing that value. The red markers plot the empirical distribution of $\|ab\|_2^2$,

while the blue markers represent $\|\mathbf{A}\mathbf{b}\|_2^2$. Thus, the blue curve corresponds to the independence-based prediction commonly assumed in existing analyses, whereas the red curve captures the actual behavior of polynomial products. Notably, the distribution of $\|ab\|_2^2$ exhibits a substantially heavier right tail, meaning that large-norm events occur far more frequently than the i.i.d. model predicts. This mismatch shows that matching one-coordinate marginals does not control the joint tail relevant to DFR.

3.2 Norm-wise Model

We now record the exact algebraic identities behind the norm-wise model. The main point is that the contribution missed by the coefficient-wise approximation is not a vague geometric effect, but a concrete cross term built from negacyclic autocorrelations.

Lemma 1 (Norm identity). *For every $a \in \mathcal{R}_n$,*

$$\|a\|_2^2 = (a\bar{a})_{[0]}. \quad (4)$$

Proof. Since $a\bar{a} = \sum_{u,v} a_{[u]}a_{[v]}x^{u-v}$, we have $(a\bar{a})_{[0]} = \sum_{u=0}^{n-1} a_{[u]}^2 = \|a\|_2^2$.

Lemma 2 (Norm decomposition). *For every $a, b \in \mathcal{R}_n$,*

$$\|ab\|_2^2 = (ab\bar{a}\bar{b})_{[0]} = \boxed{\|a\|_2^2 \|b\|_2^2} + \boxed{2\langle (a\bar{a})_{[1:n/2]}, (b\bar{b})_{[1:n/2]} \rangle}. \quad (5)$$

Proof. By Lemma 1,

$$\|ab\|_2^2 = ((ab)\overline{(ab)})_{[0]} = ((a\bar{a})(b\bar{b}))_{[0]}.$$

Because $\overline{a\bar{a}} = a\bar{a}$, the coefficients of $a\bar{a}$ in $\mathcal{R}_n$ satisfy $(a\bar{a})_{[n-k]} = -(a\bar{a})_{[k]}$ for $1 \leq k < n/2$, and $(a\bar{a})_{[n/2]} = 0$. Consequently,

$$a\bar{a} = (a\bar{a})_{[0]} + \sum_{k=1}^{n/2-1} (a\bar{a})_{[k]}(x^k - x^{n-k}),$$

and analogously for $b\bar{b}$. Extracting the coefficient of x^0 in the product gives

$$((a\bar{a})(b\bar{b}))_{[0]} = (a\bar{a})_{[0]}(b\bar{b})_{[0]} + 2 \sum_{k=1}^{n/2-1} (a\bar{a})_{[k]}(b\bar{b})_{[k]},$$

which is exactly (5).

Lemma 3 (Coefficient inner-product identity). *For every $a, b \in \mathcal{R}_n$ and every $i \in \{0, \dots, n-1\}$,*

$$(ab)_{[i]} = \langle a_{[0:n]}, (x^i \cdot \bar{b})_{[0:n]} \rangle. \quad (6)$$

Consequently, for any i, j ,

$$\langle (x^i \cdot \bar{b})_{[0:n]}, (x^j \cdot \bar{b})_{[0:n]} \rangle = (b\bar{b})_{[j-i]}. \quad (7)$$

Proof. The coefficient $(ab)_{[i]}$ is the negacyclic convolution sum $\sum_{u=0}^{n-1} a_{[u]}(x^i \bar{b})_{[u]}$, which is exactly the inner product in (6). Applying the same identity to the pair $(x^i \bar{b}, \overline{x^j \bar{b}})$ yields (7).

Equation (5) separates the diagonal energy term $\|a\|_2^2 \|b\|_2^2$ from the signed cross term $\langle (a\bar{a})_{[1:n/2]}, (b\bar{b})_{[1:n/2]} \rangle$. When the distributions of the coefficients of a and b are both centered at 0, the expectation of the covariance is 0, as shown in [41], meaning the expectation of $\langle (a\bar{a})_{[1:n/2]}, (b\bar{b})_{[1:n/2]} \rangle$ is zero. However, the fluctuations in covariance still lead to heavy-tailed behavior in $\|ab\|_2^2$, thickening the tail of the distribution and thereby increasing the DFR for schemes that rely on the independence assumption. When asymmetric error distributions are used, the dashed-box term can contribute much larger values, exacerbating tail amplification. Equation (6) also explains why inner distortion governs joint tail events. If the same vector $a_{[0:n]}$ has unusually large inner products with two shifted copies of $\bar{b}$, then those shifts must overlap substantially, and Equation (7) shows that their overlap is measured exactly by $(b\bar{b})_{[j-i]}$. This is the algebraic mechanism by which large off-zero autocorrelations make several coefficients of ab large simultaneously.

Remark 1. For the tri-cyclotomic ring defined by $\mathbb{Z}[x]/(x^n - x^{n/2} + 1)$, with $n = 2^i 3^j$, as used in certain NTRU-based schemes [46,37,40], Lemmas 1 and 2 do not hold in this form. Extending the norm-wise model to that setting requires additional analysis, along with the related analysis in [32].

Heuristic 1 (Asymptotic threshold-selection guide) *To calibrate exploratory trim thresholds, assume that the coefficients of a are modeled by an i.i.d. centered Gaussian surrogate with variance σ^2 , and further approximate the nonzero coefficients of $a\bar{a}$ by independent centered Gaussians with variance $n\sigma^4$. Under this surrogate,*

$$\begin{aligned} \mu_{\text{outer}} &:= \sigma^2 n, \quad \mu_{\text{inner}} := \sigma^2 \sqrt{n} \left(\sqrt{2 \log\left(\frac{n}{2} - 1\right)} - \frac{\log \log\left(\frac{n}{2} - 1\right) + \log \pi}{2 \sqrt{2 \log\left(\frac{n}{2} - 1\right)}} \right) \\ \sigma_{\text{outer}} &:= \sigma^2 \sqrt{2n}, \quad \sigma_{\text{inner}} := \frac{\pi \sigma^2 \sqrt{n}}{\sqrt{12 \log\left(\frac{n}{2} - 1\right)}} \end{aligned}$$

serve as heuristic location and scale parameters for the outer statistic $\|a\|_2^2$ and the inner score $\|(a\bar{a})_{[1:n/2]}\|_\infty$. We use them only for threshold selection in the exploratory experiments below; the certified DFR analysis of Section 3.3 does not rely on this surrogate model.

Motivated by Heuristic 1, we use the centered trimming predicate. A polynomial a is *centered-trimmed* (and similarly for b) when both

$$|\|a\|_2^2 - \mu_{\text{outer}}| \leq \omega_{\text{outer}} \sigma_{\text{outer}} \quad \text{and} \quad |\|(a\bar{a})_{[1:n/2]}\|_\infty - \mu_{\text{inner}}| \leq \omega_{\text{inner}} \sigma_{\text{inner}} \quad (8)$$

hold; otherwise we reject a (resp. b). A centered rule is preferable here to one-sided trimming because it perturbs the outer-energy distribution less severely; in fixed-weight samplers such as LAC, the outer term is already nearly fixed, and later sections therefore trim only the inner statistic. To isolate the convolution effect, Figure 4 uses $\omega_{\text{outer}} = +\infty$ together with $\omega_{\text{inner}} \in \{1, 2\}$. As trimming becomes stricter, the polynomial-product distribution moves closer to the unstructured baseline, confirming that the excess tail mass comes from convolution-induced dependence.

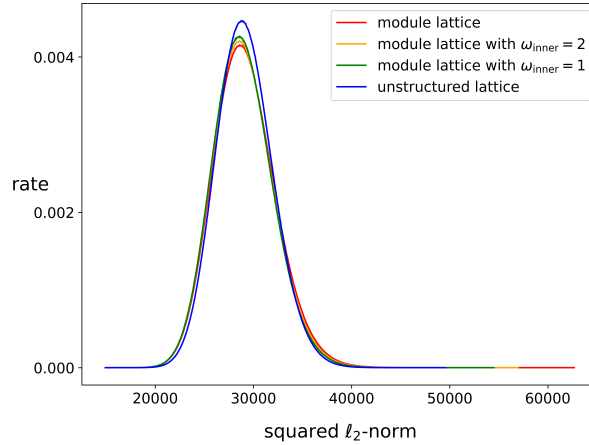


Fig. 4: Empirical effect of centered inner-distortion trimming on $\|ab\|_2^2$. Tightening the trim parameter removes the strongest dependency outliers and contracts the polynomial-product tail toward the i.i.d. matrix baseline.

3.3 From Dependency to DFR

In order to transfer the quantifiable dependency to the analysis of DFR, we show the viewpoint of *inner distortion* after canonical embedding.

Lemma 4 (Canonical embedding identity). *For every $a \in \mathcal{R}_n$ and every $k \in \{0, \dots, n-1\}$,*

$$(a\bar{a})_{[k]} = \frac{1}{n} \sum_{r=0}^{n-1} |a(\omega_r)|^2 \omega_r^{-k}. \quad (9)$$

Proof. The negacyclic Fourier transform diagonalizes multiplication in $\mathcal{R}_n$, so $(a\bar{a})(\omega_r) = a(\omega_r)\overline{a(\omega_r)} = |a(\omega_r)|^2$. Equation (9) is the inverse Fourier expansion of this power spectrum.

The vector $(a\bar{a})_{[1:n/2]}$ can be read as the collection of off-zero negacyclic auto-correlations, as the nonconstant Fourier modes of the power spectrum $|a(\omega_r)|^2$,

as the overlap profile between shifted copies of $\bar{a}$, and as the trimming score used later in the paper. Equation (9) shows that the nonzero coefficients of $a\bar{a}$ are exactly the nonconstant Fourier modes of the power spectrum $r \mapsto |a(\omega_r)|^2$. Thus, a large inner distortion means that the spectral energy of a is anisotropic across the negacyclic embedding coordinates.

3.3.1 Rigorous bound from conditional covariance control We then propose a dependency-aware DFR analysis for module-lattice-based schemes with hard-decision ECC. By following the notations in Section 2.3, we use the exact decryption-noise model

$$z = \sum_{u=1}^m a_u b_u + \eta,$$

where $A := (a_1, \dots, a_m)$ collects the left factors whose trim statistics are controlled, the polynomials b_u have independent centered coefficients with coefficient variances σ_u^2 , and η is an independent centered additive term with coefficient variance σ_η^2 . For the generic module-lattice-based encryption, one may take

$$z = \sum_{j=1}^l r_j e_j - \sum_{j=1}^k e_{1,j} s_j + e_2 \quad (10)$$

and condition on $A = (r_1, \dots, r_l, -e_{1,1}, \dots, -e_{1,k})$ by isolating the multiplicative term (e.g. $x - 1$ in NTRU-HRSS [34]) and absorbing the compression/rounding contribution into η . In particular, the variables z , Δ , t , and N_Δ are fixed before any dependence proxy is introduced. This coefficient-i.i.d. setting is the abstract regime in which the theorem below is stated. When a concrete sampler matches these hypotheses, the resulting bound is certified.

Let $\omega_r := \exp\left(\frac{(2r+1)\pi i}{n}\right)$ be the negacyclic evaluation points and let F_n denote the corresponding unitary Fourier matrix. Conditioned on A , each map $b_u \mapsto a_u b_u$ is diagonalized by F_n , and since the coefficients of b_u are independent, centered, and have covariance σ_u^2 , the conditional covariance of z takes the form

$$\text{Cov}(z \mid A) = F_n^* \text{diag}(\lambda_0(A), \dots, \lambda_{n-1}(A)) F_n, \quad \lambda_r(A) = \sigma_\eta^2 + \sum_{u=1}^m \sigma_u^2 |a_u(\omega_r)|^2.$$

For sampler classes whose right factors have independent centered coefficients, this template applies directly. For LAC, one can still use Equation (10), but because the fixed-Hamming-weight factors are not coefficient-wise independent, the latter rows of the LAC case study are reported only as conditional/empirical diagnostics rather than as direct certified corollaries of the theorem below.

Based on the link with inner distortion and canonical embedding from (9), we define

$$\Lambda(A) := \max_{0 \leq r < n} \lambda_r(A), \quad \Lambda_{\text{trim}} := \sup_{A \in \mathcal{A}_{\text{trim}}} \Lambda(A),$$

where $\mathcal{A}_{\text{trim}}$ is the set of accepted left-factor tuples.

Lemma 5 (Trim-to-spectrum control). *For every $a \in \mathcal{R}_n$,*

$$\max_{0 \leq r < n} |a(\omega_r)|^2 \leq (a\bar{a})_{[0]} + 2 \sum_{k=1}^{n/2-1} |(a\bar{a})_{[k]}|.$$

Consequently, if an accepted sample satisfies $\|(a\bar{a})_{[1:n/2]}\|_\infty \leq B_{\text{inner}}$, then

$$\max_{0 \leq r < n} |a(\omega_r)|^2 \leq \|a\|_2^2 + (n-2)B_{\text{inner}}.$$

Proof. By Lemma 4,

$$|a(\omega_r)|^2 = \sum_{k=0}^{n-1} (a\bar{a})_{[k]} \omega_r^k.$$

Taking absolute values and using $|\omega_r| = 1$ gives

$$|a(\omega_r)|^2 \leq (a\bar{a})_{[0]} + \sum_{k=1}^{n-1} |(a\bar{a})_{[k]}|.$$

Because $(a\bar{a})_{[n-k]} = -(a\bar{a})_{[k]}$ for $1 \leq k < n/2$ and $(a\bar{a})_{[n/2]} = 0$, the nonzero terms occur in pairs of equal magnitude, so

$$\sum_{k=1}^{n-1} |(a\bar{a})_{[k]}| = 2 \sum_{k=1}^{n/2-1} |(a\bar{a})_{[k]}|.$$

This proves the first claim, and the second follows from $(a\bar{a})_{[0]} = \|a\|_2^2$ together with the bound $|(a\bar{a})_{[k]}| \leq B_{\text{inner}}$ for $1 \leq k < n/2$.

Thus trimming $\|(a_u \bar{a}_u)_{[1:n/2]}\|_\infty$ suppresses the nonconstant Fourier modes of the power spectrum $r \mapsto |a_u(\omega_r)|^2$ and yields an explicit spectral envelope. For example, if each accepted a_u additionally satisfies $\|(a_u \bar{a}_u)_{[1:n/2]}\|_\infty \leq B_u$ and $\|a_u\|_2^2 \leq E_u$, then

$$\Lambda(A) \leq \sigma_\eta^2 + \sum_{u=1}^m \sigma_u^2 (E_u + (n-2)B_u). \quad (11)$$

This is the rigorous bridge from the trimming statistic to the spectral control parameter used in the certified DFR bound.

Theorem 1 (Certified DFR upper bound under trimmed spectrum).

Assume that for every accepted $A \in \mathcal{A}_{\text{trim}}$, the conditional random vector $z_{[0:n]}$ is centered and has covariance

$$\text{Cov}(z \mid A) = F_n^* \text{diag}(\lambda_0(A), \dots, \lambda_{n-1}(A)) F_n.$$

Then the trimmed worst-case DFR satisfies

$$\Pr[N_\Delta > t \mid A \in \mathcal{A}_{\text{trim}}] \leq \text{DFR}_{\text{dep,upper}} := \binom{n}{t+1} 2^{t+1} \cdot \frac{\Lambda_{\text{trim}}}{(t+1)\Delta^2}. \quad (12)$$

Proof. Fix an accepted tuple $A \in \mathcal{A}_{\text{trim}}$. If $N_\Delta > t$, then there exists a subset $S \subseteq \{0, \dots, n-1\}$ with $|S| = t+1$ such that $|z_{[i]}| > \Delta$ for every $i \in S$. For this subset choose signs $\varepsilon_i := \text{sgn}(z_{[i]})$. Then

$$\sum_{i \in S} \varepsilon_i z_{[i]} = \sum_{i \in S} |z_{[i]}| > (t+1)\Delta.$$

Hence

$$\{N_\Delta > t\} \subseteq \bigcup_{\substack{S \subseteq \{0, \dots, n-1\} \\ |S|=t+1}} \bigcup_{\varepsilon \in \{\pm 1\}^S} \left\{ \sum_{i \in S} \varepsilon_i z_{[i]} > (t+1)\Delta \right\}.$$

For fixed S and ε , set $x = \sum_{i \in S} \varepsilon_i e_i$, where e_i denotes the i -th standard basis vector of $\mathbb{R}^n$. Then $\|x\|_2^2 = t+1$, and

$$Y_{S,\varepsilon} := \sum_{i \in S} \varepsilon_i z_{[i]} = \langle x, z_{[0:n]} \rangle$$

is centered conditional on A . Moreover,

$$\text{Var}(Y_{S,\varepsilon} \mid A) = x^\top \text{Cov}(z \mid A) x \leq \Lambda(A) \|x\|_2^2 = \Lambda(A)(t+1) \leq \Lambda_{\text{trim}}(t+1).$$

Therefore, by Chebyshev's inequality,

$$\Pr[Y_{S,\varepsilon} > (t+1)\Delta \mid A] \leq \Pr[|Y_{S,\varepsilon}| > (t+1)\Delta \mid A] \leq \frac{\text{Var}(Y_{S,\varepsilon} \mid A)}{(t+1)^2 \Delta^2} \leq \frac{\Lambda_{\text{trim}}}{(t+1)\Delta^2}.$$

There are $\binom{n}{t+1}$ choices of S and 2^{t+1} sign patterns, so the union bound gives

$$\Pr[N_\Delta > t \mid A] \leq \binom{n}{t+1} 2^{t+1} \cdot \frac{\Lambda_{\text{trim}}}{(t+1)\Delta^2}.$$

Since the right-hand side is uniform over all accepted A , the same bound holds after conditioning only on the trim event $A \in \mathcal{A}_{\text{trim}}$.

Corollary 1 (Sharpened bound under conditional linear-form subgaussianity). *Assume in addition that there exists $K \geq 1$ such that for every accepted $A \in \mathcal{A}_{\text{trim}}$, every $x \in \mathbb{R}^n$, and every $\theta \in \mathbb{R}$,*

$$\mathbb{E}[\exp(\theta \langle x, z_{[0:n]} \rangle) \mid A] \leq \exp\left(\frac{K^2 \theta^2}{2} x^\top \text{Cov}(z \mid A) x\right). \quad (13)$$

Then

$$\Pr[N_\Delta > t \mid A \in \mathcal{A}_{\text{trim}}] \leq \text{DFR}_{\text{dep,upper}}^{\text{subg}} := \binom{n}{t+1} 2^{t+1} \exp\left(-\frac{(t+1)\Delta^2}{2K^2 \Lambda_{\text{trim}}}\right). \quad (14)$$

Proof. Repeat the proof of Theorem 1 up to the signed sum

$$Y_{S,\varepsilon} = \langle x, z_{[0:n]} \rangle, \quad \|x\|_2^2 = t + 1.$$

By (13), conditional on A , the random variable $Y_{S,\varepsilon}$ is $K^2 x^\top \text{Cov}(z \mid A)x$ -subgaussian. Hence the Chernoff bound gives

$$\Pr[Y_{S,\varepsilon} > (t+1)\Delta \mid A] \leq \exp\left(-\frac{(t+1)^2 \Delta^2}{2K^2 x^\top \text{Cov}(z \mid A)x}\right).$$

Using

$$x^\top \text{Cov}(z \mid A)x \leq \Lambda(A)\|x\|_2^2 = \Lambda(A)(t+1) \leq \Lambda_{\text{trim}}(t+1),$$

we obtain

$$\Pr[Y_{S,\varepsilon} > (t+1)\Delta \mid A] \leq \exp\left(-\frac{(t+1)\Delta^2}{2K^2 \Lambda_{\text{trim}}}\right).$$

A union bound over all $\binom{n}{t+1}$ choices of S and all 2^{t+1} sign patterns yields (14).

Remark 2. The hypothesis (13) is a sampler-dependent average-case concentration assumption, not a consequence of covariance control alone. It is reasonable for common lattice samplers because, conditioned on the left factors A , the noise term $z = \sum_{u=1}^m a_u b_u + \eta$ is linear in the remaining randomness (b_u, η) . When the coefficients of the b_u 's and η are independent, centered, and bounded or subgaussian, every conditional linear form $\langle x, z \rangle \mid A$ is naturally expected to be subgaussian. We emphasize, however, that this is an average-case justification: for structured samplers such as fixed-Hamming-weight distributions, (13) should still be treated as an additional hypothesis unless it is proved separately.

3.3.2 Heuristic three-vector supplement To obtain a sharper bound, we now present the three-vector model (visualized in Figure 5) as a heuristic single-coordinate approximation. It is used for calibrated interpretation.

Lemma 6 (Exact projection lemma). *Let $a, b \in \mathcal{R}_n$ and fix $i \neq 0$. Define*

$$S_0 := (ab)_{[0]}, \quad C_i := (b\bar{b})_{[i]}, \quad R_a := \|a\|_2, \quad R_b := \|b\|_2,$$

and

$$v_1 := a_{[0:n]}, \quad v_2 := \bar{b}_{[0:n]}, \quad v_3 := (x^i \bar{b})_{[0:n]}.$$

Then

$$\langle v_1, v_2 \rangle = S_0, \quad \langle v_1, v_3 \rangle = (ab)_{[i]}, \quad \langle v_2, v_3 \rangle = C_i,$$

and

$$(ab)_{[i]} = \mu_i + \delta_i, \quad \mu_i := \frac{S_0 C_i}{R_b^2}, \quad |\delta_i| \leq \sqrt{R_a^2 - \frac{S_0^2}{R_b^2}} \sqrt{R_b^2 - \frac{C_i^2}{R_b^2}}.$$

Heuristic 2 (Dependence-aware heuristic DFR estimate) Fix $i \neq 0$ and condition on b, R_a, S_0 , and C_i . In the decomposition of Theorem 6, assume that the normalized vector $w/\|w\|_2$ is approximately uniform on the unit sphere of $v_2^\perp$ and approximately independent of the direction of u_i . Then

$$(ab)_{[i]} \approx \mu_i + \sigma_i W_i, \quad \sigma_i := \sqrt{R_a^2 - \frac{S_0^2}{R_b^2}} \sqrt{R_b^2 - \frac{C_i^2}{R_b^2}},$$

where W_i has the first-coordinate law on the sphere S^{n-2} . Equivalently,

$$\frac{W_i + 1}{2} \sim \text{Beta}\left(\frac{n-2}{2}, \frac{n-2}{2}\right).$$

For $x \in [-1, 1]$, write

$$T_n(x) := \frac{1}{2} I_{1-x^2}\left(\frac{n-2}{2}, \frac{1}{2}\right) = \Pr[W_i > x],$$

and extend T_n by $T_n(x) = 1$ for $x \leq -1$ and $T_n(x) = 0$ for $x \geq 1$. Under Heuristic 2,

$$\Pr[|(ab)_{[i]}| > \Delta \mid b, R_a, S_0, C_i] \approx T_n\left(\frac{\Delta - \mu_i}{\sigma_i}\right) + T_n\left(\frac{\Delta + \mu_i}{\sigma_i}\right). \quad (15)$$

This approximation is coefficient-side and conditions on the actual shift-correlation value $C_i = (b\bar{b})_{[i]}$, rather than on the spectral envelope $\Lambda(A)$ used in the certified covariance bound. Let $E_j := \{|(ab)_{[j]}| > \Delta\}$. Under trimming, define

$$p_{\text{trim}} := \Pr(E_0 \mid \|(b\bar{b})_{[1:n/2]}\|_\infty \leq B_{\text{inner}})$$

and, for any fixed $i \neq 0$,

$$q_{\text{trim}} := \Pr(E_0 \wedge E_i \mid \|(b\bar{b})_{[1:n/2]}\|_\infty \leq B_{\text{inner}}).$$

Using (15), we approximate

$$q_{\text{trim}} \approx \mathbb{E}\left[\mathbf{1}_{E_0} \left(T_n\left(\frac{\Delta - \mu_i}{\sigma_i}\right) + T_n\left(\frac{\Delta + \mu_i}{\sigma_i}\right)\right) \mid \|(b\bar{b})_{[1:n/2]}\|_\infty \leq B_{\text{inner}}\right] \quad (16)$$

Here μ_i and σ_i are computed from the conditioned values of $b, R_a, S_0 = (ab)_{[0]}$, and $C_i = (b\bar{b})_{[i]}$. As a second heuristic step, we approximate the exceedance indicators with an exchangeable Bernoulli family that shares the same first two moments, yielding a beta-binomial model for N_Δ . Define

$$\rho_{\text{trim}} := \frac{q_{\text{trim}} - p_{\text{trim}}^2}{p_{\text{trim}}(1 - p_{\text{trim}})},$$

and, when $0 < \rho_{\text{trim}} < 1$, set

$$\alpha_{\text{trim}} := p_{\text{trim}}(\rho_{\text{trim}}^{-1} - 1), \quad \beta_{\text{trim}} := (1 - p_{\text{trim}})(\rho_{\text{trim}}^{-1} - 1).$$

This leads to the beta-binomial closure of DFR estimate as

$$\text{DFR}_{\text{dep,heur}} := \sum_{s=t+1}^n \binom{n}{s} \frac{\text{Beta}(s + \alpha_{\text{trim}}, n - s + \beta_{\text{trim}})}{\text{Beta}(\alpha_{\text{trim}}, \beta_{\text{trim}})}, \quad (17)$$

where $\text{Beta}(x, y) := \int_0^1 t^{x-1} (1-t)^{y-1} dt$ denotes the Beta function.

Proof. The three displayed inner-product identities are immediate from Equation (6). Decompose v_1 and v_3 orthogonally with respect to v_2 :

$$v_1 = \frac{S_0}{R_b^2} v_2 + w, \quad v_3 = \frac{C_i}{R_b^2} v_2 + u_i,$$

where $w, u_i \perp v_2$. Then

$$(ab)_{[i]} = \langle v_1, v_3 \rangle = \frac{S_0 C_i}{R_b^2} + \langle w, u_i \rangle.$$

This gives the claimed decomposition with $\delta_i := \langle w, u_i \rangle$. Finally,

$$\|w\|_2^2 = R_a^2 - \frac{S_0^2}{R_b^2}, \quad \|u_i\|_2^2 = R_b^2 - \frac{C_i^2}{R_b^2},$$

so the bound on $|\delta_i|$ follows from Cauchy–Schwarz.

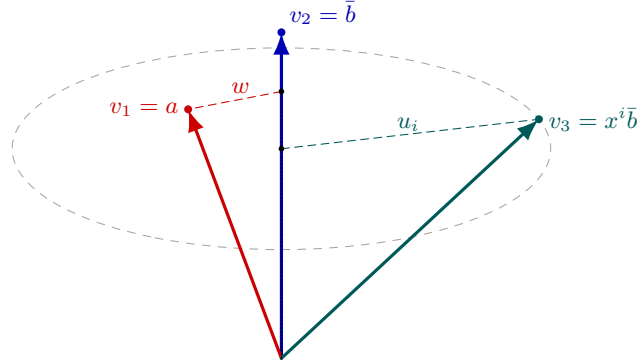


Fig. 5: Geometric visualization of the three-vector model. The axis vector is $v_2 = \bar{b}$, the vector $v_1 = a$ is close to the axis, and the shifted copy $v_3 = x^i \bar{b}$ has the same norm as v_2 and a prescribed axial correlation C_i . The dashed segments indicate the orthogonal components in the decompositions of Theorem 6.

3.3.3 Validation of the DFR estimates We now validate the three-vector heuristic from three complementary perspectives: the conditional one-coordinate law, the dependence-aware trend under trimming, and the count-level closure that converts pairwise exceedance information into a DFR estimate. Throughout this subsection, we use the parameter setting $n = 512$ and balanced fixed Hamming weight $h = 256$ with equal ± 1 counts, and we use 2^{20} samples in each experiment. We first test the core one-coordinate prediction of the three-vector model in a rare, highly dependent regime. We sample a uniformly from the fixed-weight distribution and sample b from a Type-2 family corresponding to a rare tail event under the ambient sampler. For each sample, we recenter the product ab at a trigger coefficient and normalize its sign so that the trigger coefficient becomes positive. We then condition on the slice $S_0 = (ab)_{[0]} = 34$, $C_i = (b\bar{b})_{[i]} = 5$, and examine the empirical conditional law of $(ab)_{[i]}$. Figure 6 compares this empirical law against the one-dimensional distribution predicted by the three-vector model, together with a Gaussian fit using the empirical mean and variance. The agreement is good with full visible support. This supports the intended role of the model: once the trigger value S_0 and the shift-correlation value C_i are fixed, the conditional law of a neighboring coefficient is well approximated by the spherical-slice formula. To make the comparison more intrinsic, we also study the normalized residual

$$W_i^{\text{emp}} := \frac{(ab)_{[i]} - \mu_i}{\sigma_i},$$

where μ_i and σ_i are the location and scale parameters from the exact projection lemma. Figure 7 shows that the residual is sharply concentrated near 0. Thus, in the chosen rare-event slice, the residual is more concentrated than the raw sphere-coordinate law.

Next, we test whether the heuristic captures the dependence trend induced by trimming. We sweep the trim threshold $B_{\text{inner}} \in \{28, 30, 32, 34, 36, 38, 40\}$ and compare the empirical DFR against its heuristic counterpart. To avoid introducing an additional model for the full trimmed distribution of $C_i = (b\bar{b})_{[i]}$, we use a worst-edge surrogate and place all of its mass on the trim boundary such that $|C_i| = B_{\text{inner}}$. Therefore, the three-vector formula depends only on the boundary magnitude B_{inner} , and the resulting quantity should be interpreted as an edge-of-trim dependence-aware heuristic estimate. Figure 8 compares the empirical DFR, the certified DFR upper bound $\text{DFR}_{\text{dep,upper}}$, the sharpened DFR upper bound under subgaussianity $\text{DFR}_{\text{dep,upper}}^{\text{subg}}$, and the three-vector-model heuristic DFR $\text{DFR}_{\text{dep,heur}}$. Both the empirical and heuristic DFRs increase as trimming is relaxed, indicating that the heuristic correctly predicts how the dependence effect changes with B_{inner} .

4 Exploit Dependencies in CCA Attacks against Module-Lattice Encryption with ECC

This norm-wise perspective also yields an attack-side heuristic for cryptanalysis. A key step in CCA attacks leveraging high DFR is to craft “bad” ciphertexts

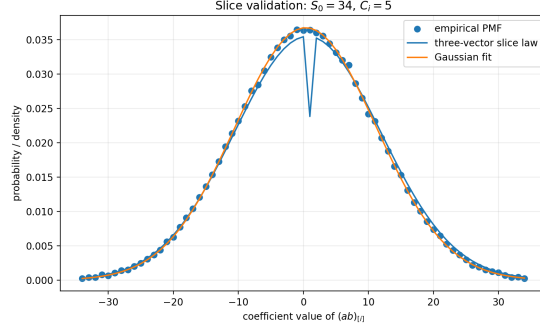


Fig. 6: Conditional slice validation for the three-vector model.

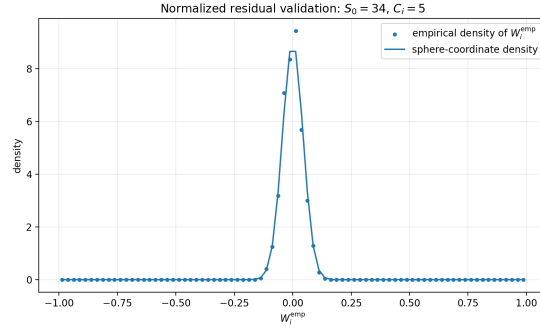


Fig. 7: Validation of the normalized residual $W_i^{\text{emp}} = ((ab)_{[i]} - \mu_i) / \sigma_i$.

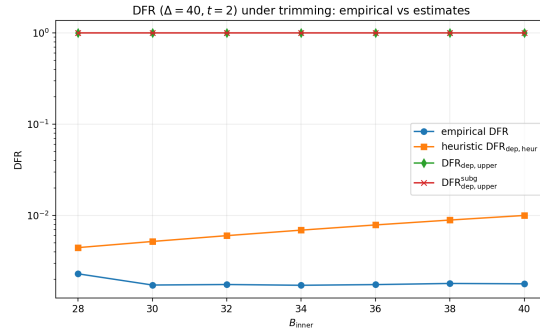


Fig. 8: Comparison among empirical data, upper bounds, and heuristic estimate.

whose failure probability is substantially above average. In the LAC setting, the attacker searches over candidate fixed-weight randomness polynomials drawn from the same sampler as the scheme and looks for instances whose depen-

dependency score predicts elevated tail behavior. Prior work describes this search using message-derived patterns because those patterns induce particular instances of randomness [29]; our point here is more generic: the relevant predictor is the resulting polynomial dependency profile, rather than a specific syntactic pattern family. The effectiveness of patterns in [29] is therefore explained by the large inner distortion they induce. Our norm-wise model makes this dependency explicit and suggests a dependency-guided proxy selector rather than a fixed-pattern heuristic.

In LAC encryption, the distributions $\mathcal{X}_r$ and $\mathcal{X}_{e_1}$ are fixed-Hamming-weight distributions, which effectively fix the *outer distortion*. From the norm-wise perspective, the relevant selector is therefore the inner-distortion score $\|(a\bar{a})_{[1:n/2]}\|_\infty$ from Section 3.2. The adversarial strategy of [29] focuses on polynomials with an interval that contains only 1 or -1 and the rest are 0s, which was called the TYPE 2 pattern, and the TYPE 1 pattern is a special case with no 0 in the interval. While such patterns do indeed yield large inner distortion, our model shows that they are merely a special case within a larger class of exploitable ciphertexts; any polynomial whose coefficients exhibit the large *inner distortion* tends to produce a heavier upper tail in the resulting product distribution.

To demonstrate that these patterns actually possess the property of high distortion empirically, we conducted a simple experiment in $\mathcal{R}_{512}$, using the LAC NIST-I distribution: 128 coefficients equal to 1, 128 equal to -1 , and the remainder equal to 0. We first sampled a baseline pool of 2^{16} random polynomials a from this distribution, a pool of 2^{16} random polynomials b_1 from the TYPE 1 distribution with 37 consecutive 1, and a pool of 2^{16} random polynomials b_2 from the TYPE 2 distribution with an interval that contains 60 positive 1 (no need to be consecutive) and 42 zeros. The two patterns occur with comparable probabilities, $2^{-65.41}$ and $2^{-65.35}$ respectively, computed via the formula in Section 4 of [29]. We then compute $\|(a\bar{a})_{[1:n/2]}\|_2$ and $\|(a\bar{a})_{[1:n/2]}\|_\infty$ for each pool. Figure 9 reports the result.

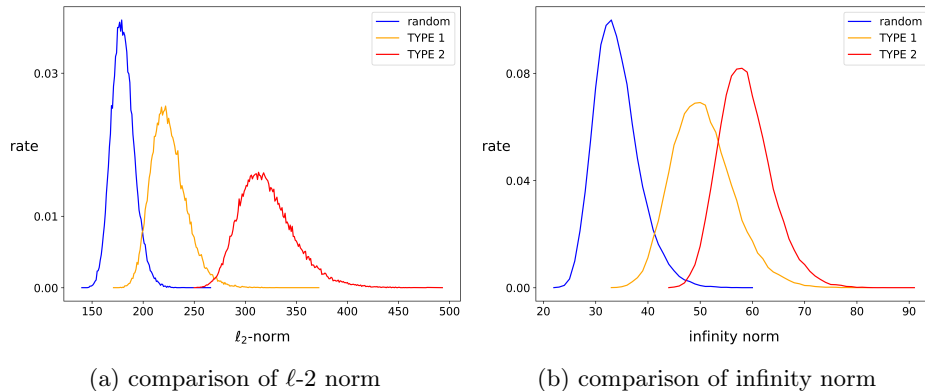


Fig. 9: Comparison of inner distortion among different patterns.

The figure shows that TYPE 2 induces strictly larger inner-distortion norms than TYPE 1, which in turn dominates the uniform baseline. As noted in [29], TYPE 2 performs better than TYPE 1 at the same probability; this figure directly supports this. The *inner distortion* serves as a metric to evaluate the quality of the pattern, which implies that we could use this metric to find more patterns with more effects on decryption failure. This also implies a heuristic: when we set a threshold for *inner distortion*, the pattern with higher DFR would be accepted with negligible probability.

Moreover, to illustrate the benefits of the *inner distortion* metric check rather than the specific pattern check. We conduct another simple experiment in $\mathcal{R}_{512}$, still using the LAC NIST-I sampler. We first fix 100 random polynomials a from the distribution with fixed hamming weight 256 as the secret keys, and then scan an additional 2^{18} candidates under each of the two selection criteria below:

- *Inner-distortion selector*: retain polynomials $b \in \mathcal{R}_{512}$ satisfying

$$\|(b\bar{b})_{[1:n/2]}\|_{\infty} \geq 40.$$

- *Single-signal interval selector* (TYPE 2): retain polynomials $c \in \mathcal{R}_{512}$ containing an interval with at least 11 positive entries or 11 negative entries, and the rest are at most 19 zeros.

By following the notations in Section 2.3, we calculate N_{Δ} with the threshold $\Delta = 40$ for each polynomial product ab and ac and treat $N_{\Delta} \geq 3$ as the ECC failure event. These are empirical tail probes chosen to yield measurable variation within feasible Monte Carlo budgets; they are not the LAC decoding rule.

The cutoffs 40 and (11, 19) are empirical rare-event selectors chosen only for this comparison. After the same search budget of $100 \cdot 2^{18}$ trials, the inner-distortion selector retained $\approx 2^{19.42}$ candidates and produced 26 ECC failure events; the continuous-run selector retained $\approx 2^{19.10}$ candidates and produced 16. The corresponding per-sample failure rates are $26/2^{19.42} \approx 3.8 \times 10^{-5}$ and $16/2^{19.10} \approx 2.9 \times 10^{-5}$, i.e., the inner-distortion selector identifies failures roughly 30% more efficiently than the pattern proxy at matched budget. We take this as evidence that the norm-wise score is a useful failure-finding heuristic.

5 Trim on Module Lattice Encryptions and Encapsulations

Given the weaknesses introduced by dependencies, in this section we present a generic, lightweight safeguard procedure for module-lattice-based public key encryption (PKE) and key encapsulation mechanisms (KEM). The central idea is to reject polynomials whose dependence statistics would otherwise lead to more severe tail behavior. Outer-distortion screening is already familiar in lattice cryptography, for example, in Falcon key generation [23] and BAT-style rejection [24], where public predicates filter large-norm outliers. We separate that familiar outer screening from the identified dependency statistics. We bridge the

dependency to DFR via the canonical embedding, and then derive a rigorous DFR upper bound from a bounded spectral envelope, and complement it with a tighter heuristic estimate driven by bounded *inner distortion*. Furthermore, given a module-lattice-based PKE scheme with algorithms **KeyGen**, **Encrypt**, and **Decrypt**, we first introduce the generic protection framework **TrimPKE**, then analyze its impact on concrete security, and finally describe techniques for efficient implementation. By applying the Fujisaki-Okamoto transform to **TrimPKE**, we obtain the corresponding **TrimKEM**. The full security proofs of **TrimPKE** and **TrimKEM** are provided in Section 6. The framework keeps the arithmetic parameters unchanged, but it introduces explicit rejection rates, runtime overhead, and support-loss terms that must be accounted for in the concrete security analysis. We instantiate it in LAC [45] and DAWN [42] as case studies.

5.1 TrimPKE

5.1.1 Framework Description We begin by introducing an auxiliary algorithm **IsDistorted**, which arises from Equation (8) and evaluates both the *inner distortion* and *outer distortion* of a polynomial. Based on this evaluation, we define two rejection-sampling procedures: $\text{Reject}_{\text{key}}$ and $\text{Reject}_{\text{enc}}$. Figure 11 shows the variant tailored for Module-LWE-based encryptions, and Figure 12 shows the variant designed for NTRU-based encryptions. The notation is consistent with the framework in Figures 1 and 2. Let ρ_0 denote the initial seed used to generate encryption randomness, and let $\mathbf{E}$ be a variable-length-input hash function modeled as a random oracle in the security analysis. A more natural choice would be to resample a fresh seed ρ_0 on each retry. However, in PKE-to-KEM constructions (KEMs, KE, AKE), the ciphertext must remain a deterministic function of the input seed ρ_0 . We therefore derive each retry’s randomness from ρ_0 via the random oracle, so that identical ρ_0 always yields identical ciphertexts.

Fig. 10: **Generic TrimPKE Framework**

TrimGen	TrimEnc(pk, m, ρ_0)
1 : $(pk, sk) \leftarrow \text{KeyGen}$	1 : $N \leftarrow 0$
2 : if $\text{Reject}_{\text{key}}(pk, sk) = 0$	2 : $\rho \leftarrow \rho_0$
3 : go back to step 1	3 : $\rho \leftarrow \mathbf{E}(N, \rho \rho_0)$
4 : return (pk, sk)	4 : $c \leftarrow \text{Encrypt}(pk, m, \rho)$
	5 : if $\text{Reject}_{\text{enc}}(pk, m, \rho, c) = 0$
TrimDec(c, sk)	6 : $N \leftarrow N + 1$
1 : $m \leftarrow \text{Decrypt}(c, sk)$	7 : go back to step 3
2 : return m	8 : return c

We denote by $\sigma_s, \sigma_e, \sigma_r, \sigma_{e_1}, \sigma_g, \sigma_f, \sigma_t$ the standard deviations of the coefficient distributions $\mathcal{X}_s, \mathcal{X}_e, \mathcal{X}_r, \mathcal{X}_{e_1}, \mathcal{X}_g, \mathcal{X}_f, \mathcal{X}_t$ of the corresponding polynomials. We align $\omega_{\text{outer}}, \omega_{\text{inner}}$ into one single value, and denote by $\omega_s, \omega_e, \omega_r, \omega_{e_1}, \omega_g, \omega_f, \omega_t$ the trim parameters of polynomials. For each sampler, let $\theta_a = (\mu_{\text{outer}}^{(a)}, \sigma_{\text{outer}}^{(a)}, \mu_{\text{inner}}^{(a)}, \sigma_{\text{inner}}^{(a)})$ denote the calibration tuple used by the predicate. The Gaussian-surrogate formulas of Section 3.2 provide one possible choice for θ_a , but concrete instantiations may instead use exact or empirical sampler moments. Let $\delta_s, \delta_e, \delta_r, \delta_{e_1}, \delta_g, \delta_f, \delta_t$ denote the pass probabilities on these calibrated predicates, where $\delta_s := \Pr_{a \leftarrow \mathcal{X}_s^n} [\text{IsDistorted}((a), \theta_s, \omega_s) = 0]$ denotes the single-polynomial pass probability; for k -dimensional vectors with independent components, the pass probability is δ_s^k .

Algorithm 1 IsDistorted

Input: vector $\mathbf{a} \in \mathcal{R}_n^k$, calibration tuple $\theta_a = (\mu_{\text{outer}}, \sigma_{\text{outer}}, \mu_{\text{inner}}, \sigma_{\text{inner}})$, trim parameter ω
Output: distorted or not $b \in \{0, 1\}$
1: $b \leftarrow 0$
2: **for** i **from** 0 **to** $k - 1$
3: **if** $|\|\mathbf{a}_{[i]}\|_2^2 - \mu_{\text{outer}}| > \omega \cdot \sigma_{\text{outer}}$ **or** $|\|(\mathbf{a}_{[i]} \bar{\mathbf{a}}_{[i]})_{[1:n/2]}\|_\infty - \mu_{\text{inner}}| > \omega \cdot \sigma_{\text{inner}}$
4: $b \leftarrow 1$
5: **return** b

We define the probabilities that $\text{Reject}_{\text{key}}$ and $\text{Reject}_{\text{enc}}$ output 0 as r_{key} and r_{enc} , respectively. For Module-LWE-based rejection, we obtain

$$\begin{aligned} r_{\text{key}} &= 1 - \delta_s^k \cdot \delta_e^l, \\ r_{\text{enc}} &= 1 - \delta_r^l \cdot \delta_{e_1}^k. \end{aligned} \tag{18}$$

For NTRU-based rejection, we have

$$r_{\text{key}} = 1 - \delta_g \cdot \delta_f, \quad r_{\text{enc}} = 1 - \delta_t. \tag{19}$$

We include (pk, m, c) in the formal interface of $\text{Reject}_{\text{enc}}$ to accommodate compression-aware variants, such as Kyber [6] and DAWN [42], where the public ciphertext and public inputs may be needed to inspect a polynomial created by compression. The pass-rate formulas in Equations (18) and (19) refer to the basic setting discussed above, where the accept/reject decision depends only on the sampled polynomials. Concrete compressed instantiations may add extra public-input checks without changing the surrounding reduction interface.

5.1.2 Impact on Concrete Security We now isolate the concrete-security effect that is actually analyzed in this paper. Our rejection sampling is a public function of public parameters, which is analogous to other public-predicate rejection rules used in lattice cryptography, e.g. key generation in Falcon [23]

and BAT [24]. The timing and other side information are independent of secret key and randomness. Therefore, the concrete cryptographic effect of trimming is therefore not leakage, but a contraction of the effective sample space, which must be reflected in the hardness assumptions and concrete-security bounds. We summarize the main assumptions underlying current module-lattice-based encryption schemes as follows: Ring-LWE [45], Module-LWE [6], Ring-LWR [24], Module-LWR [16], and NTRU [30]. While these problems differ in algebraic structure, their state-of-the-art concrete-security estimates are broadly comparable, and our security analysis accounts explicitly for the loss induced by rejection.

Fig. 11: **Module-LWE-based Rejection**

$\text{Reject}_{\text{key}}^{\text{MLWE}}(pk := (\mathbf{A}, \mathbf{b}), sk := \mathbf{s})$	$\text{Reject}_{\text{enc}}^{\text{MLWE}}(pk, m, \rho := (\rho_r, \rho_{e_1}), c)$
1 : $rej \leftarrow 1$	1 : $\mathbf{r} \leftarrow \text{Sample}((\mathcal{X}_r^n)^l, \rho_r)$
2 : if $\text{IsDistorted}(\mathbf{s}, \theta_s, \omega_s) = 1$	2 : $\mathbf{e}_1 \leftarrow \text{Sample}((\mathcal{X}_{e_1}^n)^k, \rho_{e_1})$
3 : $rej \leftarrow 0$	3 : $rej \leftarrow 1$
4 : $\mathbf{e} \leftarrow \mathbf{b} - \mathbf{s}\mathbf{A}^\top \pmod{q}$	4 : if $\text{IsDistorted}(\mathbf{r}, \theta_r, \omega_r) = 1$
5 : if $\text{IsDistorted}(\mathbf{e}, \theta_e, \omega_e) = 1$	5 : $rej \leftarrow 0$
6 : $rej \leftarrow 0$	6 : if $\text{IsDistorted}(\mathbf{e}_1, \theta_{e_1}, \omega_{e_1}) = 1$
7 : return rej	7 : $rej \leftarrow 0$
	8 : return rej

Fig. 12: **NTRU-based Rejection**

$\text{Reject}_{\text{key}}^{\text{NTRU}}(pk := h, sk := f)$	$\text{Reject}_{\text{enc}}^{\text{NTRU}}(pk, m, \rho, c)$
1 : $g \leftarrow hf \pmod{q}$	1 : $t \leftarrow \text{Sample}(\mathcal{X}_t^n, \rho)$
2 : $rej \leftarrow 1$	2 : $rej \leftarrow 1$
3 : if $\text{IsDistorted}(g, \theta_g, \omega_g) = 1$	3 : if $\text{IsDistorted}(t, \theta_t, \omega_t) = 1$
4 : $rej \leftarrow 0$	4 : $rej \leftarrow 0$
5 : $f' \leftarrow (f - 1)/p$	5 : return rej
6 : if $\text{IsDistorted}(f', \theta_f, \omega_f) = 1$	
7 : $rej \leftarrow 0$	
8 : return rej	

For the power-of-two cyclotomic settings and parameter ranges used in practice, the best-known attacks against Ring/Module-LWE and Ring/Module-LWR remain lattice-reduction-based and are typically estimated using the same BKZ-style cost models as for LWE. Likewise, no attack is known that gains an advantage from the rounding operation in Ring-LWR or Module-LWR. NTRU instances are closely related structured lattice problems except for the dense-sublattice attack [17]; modern parameter sets account for this in their security estimates. However, this attack primarily affects parameter settings with large moduli; modern NTRU-based schemes account for this in their parameter choices and therefore effectively mitigate the threat. Consequently, most schemes treat Ring-LWE, Module-LWE, Ring-LWR, Module-LWR, and NTRU as instances of LWE and rely on the lattice estimator¹ from [1] to evaluate the security of the corresponding LWE or NTRU instance, which is then taken as the concrete security level of the scheme.

In what follows, we review the mainstream methods for solving LWE or NTRU and explain how our rejection sampling technique affects concrete security through explicit support-loss terms and, empirically, much smaller estimator shifts. For a LWE instance $\text{LWE}_{m,n,q,\mathcal{X}_s,\mathcal{X}_e}$, we have $\mathbf{s}\mathbf{A}^\top + \mathbf{e} \equiv \mathbf{b} \pmod{q}$, where $\mathbf{A} \in \mathbb{Z}^{m \times n}$, and the entries of $\mathbf{A}$ are drawn from the uniform distribution $\mathcal{U}(\mathbb{Z}_q)$, and the entries of $\mathbf{s}, \mathbf{e}$ are drawn from the distributions $\mathcal{X}_s$ and $\mathcal{X}_e$ whose the standard deviations are σ_s, σ_e , respectively. The mainstream methods for solving LWE are lattice-based attacks, which can be broadly categorized into two types: primal and dual attacks.

Primal Attack. To solve LWE, we can construct the lattice

$$\mathcal{L}_{\mathbf{A},q} = \mathbb{Z}^{(m+n) \times (m+n)} \cdot \begin{pmatrix} \mathbf{A}^\top & \mathbf{I}_n \\ q \cdot \mathbf{I}_m & \mathbf{0} \end{pmatrix}.$$

For a fixed target vector $\mathbf{v} = (\mathbf{b}, \mathbf{0})$, there exists a nearby lattice vector $\mathbf{v} - (\mathbf{e}, -\mathbf{s})$, with gap vector $(\mathbf{e}, -\mathbf{s})$, which is extremely short. The distribution of $\mathcal{X}_s, \mathcal{X}_e$ guarantees that $\mathbf{v}$ is the unique closest vector with overwhelming probability. Thus, the problem reduces to solving a Bounded Distance Decoding (BDD) instance over $\mathcal{L}_{\mathbf{A},q}$. Depending on the method, the primal attack has two main branches: primal BDD and primal unique SVP (uSVP).

- For primal BDD, we apply BKZ reduction [48,49] to obtain a reduced basis of $\mathcal{L}_{\mathbf{A},q}$, and then solve the BDD instance using Babai’s rounding or nearest-plane algorithms.
- For primal uSVP, we use the embedding technique [36,3] and build the extended lattice:

$$\mathcal{L}_{\mathbf{A},\mathbf{b},q} = \mathbb{Z}^{(m+n+1) \times (m+n+1)} \cdot \begin{pmatrix} \mathbf{A}^\top & \mathbf{I}_n & \mathbf{0} \\ q \cdot \mathbf{I}_m & \mathbf{0} & \mathbf{0} \\ \mathbf{b} & \mathbf{0} & 1 \end{pmatrix},$$

and then apply lattice reduction directly to solve the unique SVP instance, thereby recovering $\mathbf{s}$ and $\mathbf{e}$.

¹ <https://github.com/malb/lattice-estimator>

Dual Attack. Alternatively, we partition $\mathbf{A}$ into two submatrices $\mathbf{A}_0 \in \mathbb{Z}^{m \times n_0}$ and $\mathbf{A}_1 \in \mathbb{Z}^{m \times n_1}$, and split $\mathbf{s}$ into $\mathbf{s}_0 \in \mathbb{Z}^{n_0}$ and $\mathbf{s}_1 \in \mathbb{Z}^{n_1}$ with $n_0 + n_1 = n$. Thus,

$$\mathbf{s}_0 \mathbf{A}_0^\top + \mathbf{s}_1 \mathbf{A}_1^\top + \mathbf{e} \equiv \mathbf{b} \pmod{q}.$$

We construct the lattice as:

$$\mathcal{L}_{\mathbf{A}, q, n_0} = \mathbb{Z}^{(m+n_0) \times (m+n_0)} \cdot \begin{pmatrix} \mathbf{A}_0 & \mathbf{I}_m \\ q \cdot \mathbf{I}_{n_0} & \mathbf{0} \end{pmatrix}.$$

By applying lattice reduction to $\mathcal{L}_{\mathbf{A}, q, n_0}$, we obtain multiple short vectors $(\mathbf{w}, \mathbf{u})$ satisfying $\mathbf{u} \mathbf{A}_0 \equiv \mathbf{w} \pmod{q}$. This leads to

$$\mathbf{b} \mathbf{u}^\top - \mathbf{s}_1 \mathbf{A}_1 \mathbf{u}^\top \equiv \mathbf{s}_0 \mathbf{w}^\top + \mathbf{e} \mathbf{u}^\top \pmod{q}.$$

Since both $\mathbf{s}_0 \mathbf{w}^\top$ and $\mathbf{e} \mathbf{u}^\top$ are small, we can exhaust over possible values of $\mathbf{s}_1$, and then iteratively recover the full secret $\mathbf{s}$.

From the above, the cost of attacks depends mainly on n , q , and the standard deviations σ_s, σ_e . Trimming changes this picture in two conceptually different ways. First, it changes the output distribution by shrinking the support, which is captured information-theoretically by the pass probabilities $\delta_s, \delta_e, \delta_r, \delta_{e_1}, \delta_g, \delta_f, \delta_t$ of `lsDistorted`. Second, it may shift the effective standard deviations seen by current lattice estimators. We observed only small estimator shifts in exploratory runs, but we do not rely on a particular percentage here and keep the formal discussion at the support-loss level. Beyond pure lattice algorithms, state-of-the-art attacks also combine lattice reduction with combinatorial techniques, such as Meet-in-the-Middle (MITM). The complexity of such hybrid attacks depends on the effective search space for $\mathbf{s}$ and $\mathbf{e}$. Accordingly, for a Module-LWE-based encryption scheme, the support reduction alone can weaken the concrete security of the private key by at most $l \cdot \log_2(\delta_e^{-1}) + k \cdot \log_2(\delta_s^{-1})$ bits, while the security of the message can be weakened by at most $l \cdot \log_2(\delta_r^{-1}) + k \cdot \log_2(\delta_{e_1}^{-1})$ bits. For an NTRU-based encryption scheme, the corresponding support-loss terms are at most $\log_2(\delta_g^{-1}) + \log_2(\delta_f^{-1})$ bits for the private key and $\log_2(\delta_t^{-1})$ bits for the message. For the instantiations of TrimLAC and TrimDAWN reported below, every per-polynomial pass probability δ satisfies $\delta \geq \frac{1}{2}$, so the support-loss term is bounded by 2 bits at every NIST security level.

It is worth noting that for many parameter sets, hybrid attacks with MITM are not optimal; thus, the above constitutes only a worst-case analysis. Moreover, our rejection sampling technique filters out polynomials with abnormally small norms, thereby strengthening concrete security in specific scenarios.

5.1.3 Implementation and Acceleration The core component of our framework is the algorithm `lsDistorted`, which requires computing the product of a polynomial a with its conjugate $\bar{a}$. To achieve efficiency, we employ the Number Theoretic Transform (NTT), reducing polynomial multiplication in $\mathcal{R}_n / Q\mathcal{R}_n$ from $\mathcal{O}(n^2)$ to $\mathcal{O}(n \log n)$.

Let n be a power of two and ζ a primitive $2n$ -th root of unity. Then

$$\mathbb{Z}_Q[x]/(x^n + 1) \cong \mathbb{Z}_Q[x]/(x^{\frac{n}{2}} - \zeta^{\frac{n}{2}}) \times \mathbb{Z}_Q[x]/(x^{\frac{n}{2}} + \zeta^{\frac{n}{2}}).$$

If $\mathbb{Z}_Q$ lacks such a primitive root, we instead use a $\frac{2n}{d}$ -th root, which yields $\frac{n}{d}$ irreducible factors of degree d . The transform is *complete* when $d = 1$ and *incomplete* when $d > 1$. Moreover, the NTT of $\bar{a}$ can be derived directly from a , even in the incomplete case. Specifically, we have

$$a \equiv \sum_{i=0}^{d-1} a_{[i]} x^i \pmod{x^d - \zeta}, \quad \bar{a} \equiv \sum_{i=0}^{d-1} \zeta a_{[i]} x^{d-1-i} \pmod{x^d - \zeta^{-1}}.$$

This property eliminates one full NTT computation, further improving efficiency. We note that the standard NTT only calculates the result in $\mathbb{Z}_Q$, and we need the result of $a\bar{a}$ in $\mathbb{Z}$. However, we can use the NTT lifting techniques in [13, 43], which use a large enough modulus Q to ensure that all the coefficients in the polynomial product would not exceed $\lfloor \frac{Q}{2} \rfloor$ or with a negligible probability.

5.2 TrimKEM

By applying the FO transformation to TrimPKE, we obtain a generic TrimKEM framework as shown in Figure 13, which can be used to construct Trim-protected KEM schemes.

Fig. 13: Generic TrimKEM Framework

TrimGen ^{ℓ}	TrimDeca($c, sk = (sk', pk, H(pk), \bar{K}')$)
1 : $(pk, sk') \leftarrow \text{TrimGen}$	1 : $m' \leftarrow \text{TrimDec}(c, sk')$
2 : $\bar{K}' \leftarrow \text{Sample}(\mathcal{M})$	2 : $(\bar{K}, \rho) \leftarrow G(m', H(pk))$
3 : $sk \leftarrow (sk', pk, H(pk), \bar{K}')$	3 : if $c \neq \text{TrimEnc}(pk, m', \rho)$
4 : return (pk, sk)	4 : $K \leftarrow F(\bar{K}', c)$
	5 : else
	6 : $K \leftarrow F(\bar{K}, c)$
	7 : return K
TrimEnca(pk)	
1 : $m \leftarrow \text{Sample}(\mathcal{M})$	
2 : $(\bar{K}, \rho_0) \leftarrow G(m, H(pk))$	
3 : $c \leftarrow \text{TrimEnc}(pk, m, \rho_0)$	
4 : $K \leftarrow F(\bar{K}, c)$	
5 : return (c, K)	

5.3 TrimLAC and TrimDAWN

Because Module-LWE-based and NTRU-based schemes differ in how trimming is most naturally integrated, we present two concrete instantiations of the TrimPKE and TrimKEM frameworks. For the Module-/Ring-LWE side, we construct a trimmed variant of LAC-v3a in [45], denoted TrimLAC. For the NTRU side, we construct a trimmed variant of DAWN- α in [42], denoted TrimDAWN. (Strictly, LAC is Ring-LWE-based; we treat it as a degree-one Module-LWE instance, and its use of ECC makes it a representative testbed for our framework.)

For TrimLAC, we follow the latest reference implementation in the repository². LAC samples from fixed-Hamming-weight distributions, which already impose an implicit rejection on *outer* distortion. Accordingly, in TrimLAC we focus trimming on *inner* distortion only, by enforcing an explicit upper bound B_{inner} on $\|(a\bar{a})_{[1:n/2]}\|_{\infty}$ (and analogously for b). Since the outer behavior is effectively fixed by the sampling rule, this avoids the need to additionally control skewness in the squared ℓ_2 -norm. For this reason, in Algorithms 2-3 we do not invoke the generic predicate `IsDistorted`; instead, we directly compute and check $\|(a\bar{a})_{[1:n/2]}\|_{\infty}$. Complete algorithms for both TrimLAC and TrimDAWN are provided in Appendix A. For the DFR discussion, the only point we use is that the reseeding procedure is a public deterministic transform of the seed and public inputs. Accordingly, the case-study analysis below reports the measured rejection rates and the resulting tail metrics under the implemented resampling rule, rather than relying on a separate hash-uniformity theorem.

5.3.1 Performance Comparison We report performance and dependency evaluations of TrimLAC and TrimDAWN under varying trim parameters in Table 1. For TrimLAC, trimming is parameterized by an upper bound B_{inner} on the inner-distortion infinity norm; for TrimDAWN, the reported entries use the instantiated predicate `IsDistorted`($\cdot, \theta_{\bullet}, \omega$) with scheme-specific calibration tuples $\theta_{\bullet}$ and a scalar parameter ω . The detailed data is shown in Table 1.

To quantify the residual tail risk after trimming dependency outliers, we report the logarithmic heuristic estimate with bounded *inner distortion* $\text{DFR}_{\text{dep,heur}}$ from Equation (17), since the rigorous upper bound here would be too large such that with no reference value. We report runtime costs as tuples in keycycles. For example, (27, 46, 62) denotes the keycycle costs of key generation, encapsulation, and decapsulation for TrimLAC at NIST-I with no trimming. We also report the trim bounds B_{inner} as tuples across security levels. For example, (36, 36, 40) denotes the values of B_{inner} used at NIST-I, III, and V, respectively. To compare with the baseline without trimming, we select $B_{\text{inner}} = \infty$ (resp. ω) obtained from dependency events whose empirical probability is at least 2^{-16} . This restriction is conservative in the following sense: an adversary may invest additional offline effort to search for rarer ciphertexts with stronger dependencies, in which case trimming can provide an even larger improvement against worst-case ciphertext selection than what this baseline proxy captures.

² <https://github.com/pqc-lac/lac-intel64>

Table 1: Performance, baseline dependency-tail proxies, and instantiated inner-distortion-control metrics for TrimLAC and TrimDAWN.

Trim B_{inner}	Cost and logarithmic DFR	TrimLAC			Trim ω	TrimDAWN	
		NIST-I	NIST-III	NIST-V		NIST-I	NIST-V
(60, 44, 67)	Cost	(27, 46, 61)	(41, 69, 99)	(58, 105, 186)	9	(23, 20, 20)	(46, 36, 38)
	$\log_2(\text{DFR}_{\text{dep,heur}})$	-19	-91	-26		-103	-138
(36, 32, 40)	Cost	(34, 57, 75)	(54, 78, 115)	(79, 136, 225)	5	(25, 26, 27)	(54, 50, 61)
	$\log_2(\text{DFR}_{\text{dep,heur}})$	-31	-104	-47		-110	-144
(32, 28, 36)	Cost	(48, 91, 109)	(58, 83, 121)	(116, 229, 325)	2	(26, 28, 29)	(59, 55, 64)
	$\log_2(\text{DFR}_{\text{dep,heur}})$	-33	-108	-52		-115	-148
(28, 24, 32)	Cost	(185, 1295, 1311)	(85, 160, 192)	(547, 6121, 6131)	1	(30, 52, 45)	(79, 111, 104)
	$\log_2(\text{DFR}_{\text{dep,heur}})$	-36	-112	-58		-117	-150

6 Security Proofs of TrimPKE and TrimKEM

In this section, based on the security of the public-key encryption scheme PKE, we prove the QROM security of the previously defined TrimPKE and TrimKEM. We use $\mathcal{T}_{\mathcal{A}}$ to denote the time cost of the algorithm $\mathcal{A}$.

6.1 OW-CPA Security Proof of TrimPKE in the QROM

Theorem 2. *Let $\mathcal{A}$ be an OW-CPA adversary against TrimPKE, making q_E quantum queries to the random oracle E . Then we can construct an OW-CPA adversary $\mathcal{B}$ against PKE such that $\mathcal{T}_{\mathcal{B}} \approx \mathcal{T}_{\mathcal{A}}$ and*

$$\text{Adv}_{\text{TrimPKE}, \mathcal{A}}^{\text{OW-CPA}} \leq st \cdot \text{Adv}_{\text{PKE}, \mathcal{B}}^{\text{OW-CPA}} + (st + s) \cdot (r_{\text{enc}})^t + (s + 1) \cdot (r_{\text{key}})^s + \frac{2q_E}{\sqrt{|\mathcal{R}|}}$$

holds for any positive integers s and t . Here r_{key} and r_{enc} are defined in Eq. (18), and $\mathcal{R}$ is the randomness space of PKE.

Before giving the proof, we first point out that the upper bound given by this theorem holds for all positive integers s and t ; thus, our bound remains reasonable as long as both r_{enc} and r_{key} are relatively small. One may wonder why the bound here holds for all positive integers s and t . Taking s as an example, intuitively, our proof idea is as follows: first, we set the upper bound on the number of rejection sampling of the key generation algorithm to s ; then, we uniformly select a (pk, sk) pair from at most s such pairs, and guess that this pair is the one used for the challenge ciphertext. It can be observed that this process is independent of the specific value of s , and it only needs s to be a positive integer. In fact, the larger s is, the lower $(r_{\text{key}})^s$ becomes; however, the success probability of the random guess we make also decreases, i.e., the value of " $st \cdot \text{Adv}_{\text{PKE}, \mathcal{B}}^{\text{OW-CPA}}$ " increases. Thus, the values of s and t can be appropriately chosen based on the magnitudes of r_{key} and r_{enc} .

Proof. We introduce a sequence of games $\mathbf{G}_0$ to $\mathbf{G}_{11}$ as shown in Figure 14 and Figure 15 to prove this theorem, and the game $\mathbf{G}_0$ is the OW-CPA game of TrimPKE. Hence we have

$$\text{Adv}_{\text{TrimPKE}, \mathcal{A}}^{\text{OW-CPA}} = \Pr[1 \leftarrow \mathbf{G}_0]. \quad (20)$$

GAMES $\mathbf{G_0-G_6}$	TrimGen ₁ // $\mathbf{G_2}$	TrimGen ₃ // $\mathbf{G_4}$
1 : $(pk, sk) \leftarrow \text{TrimKeyGen}$ // $\mathbf{G_0-G_1}$ $(pk, sk) \leftarrow \text{TrimGen}_{i-1}$ // $\mathbf{G_{2 \leq i \leq 5}}$ $(pk, sk) \leftarrow \text{KeyGen}$ // $\mathbf{G_6}$ 2 : $m^* \leftarrow_{\$} \mathcal{M}, \rho_0 \leftarrow_{\$} \mathcal{R}$ // $\mathbf{G_0-G_6}$ $c^* \leftarrow \text{TrimEnc}(pk, m^*, \rho_0)$ // $\mathbf{G_0}$ $c^* \leftarrow \text{TrimEnc}_1(pk, m^*)$ // $\mathbf{G_1-G_6}$ 3 : $m \leftarrow \mathcal{A}^E(pk, c^*)$ // $\mathbf{G_0-G_5}$ $m \leftarrow \mathcal{B}(pk, c^*)$ // $\mathbf{G_6}$ 4 : return $[m = m^*]$ // $\mathbf{G_0-G_6}$	1 : $(pk, sk) \leftarrow \text{KeyGen}, n = 1$ 2 : while $\text{Reject}_{key}(pk, sk) = 0$ $\wedge n < s$ $n = n + 1$ $(pk, sk) \leftarrow \text{KeyGen}$ end while 3 : if $\text{Reject}_{key}(pk, sk) = 0$, abort game return 0 else return (pk, sk)	1 : for k from 1 to s $(pk_k, sk_k) \leftarrow \text{KeyGen}$ end for 2 : if Bad occurs, abort game return 0 else $l \leftarrow_{\$} [s]$ return (pk_l, sk_l)
TrimEnc ₁ (pk, m) // $\mathbf{G_1-G_6}$	TrimGen ₂ // $\mathbf{G_3}$	TrimGen ₄ // $\mathbf{G_5}$
1 : $\rho \leftarrow_{\$} \mathcal{R}, c \leftarrow \text{Encrypt}(pk, m, \rho)$ 2 : while $\text{Reject}_{enc}(pk, m, \rho, c) = 0$ $\rho \leftarrow_{\$} \mathcal{R}$ $c \leftarrow \text{Encrypt}(pk, m, \rho)$ end while 3 : return c	1 : for k from 1 to s $(pk_k, sk_k) \leftarrow \text{KeyGen}$ end for 2 : if Bad occurs, abort game return 0 else return (pk_i, sk_i)	1 : for k from 1 to s $(pk_k, sk_k) \leftarrow \text{KeyGen}$ end for 2 : $l \leftarrow_{\$} [s]$ return (pk_l, sk_l)

Fig. 14: Games $\mathbf{G_0}$ to $\mathbf{G_6}$ in the proof of Theorem 2.

Game $\mathbf{G_1}$: This game is the same as game $\mathbf{G_0}$, except that the encryption algorithm TrimEnc is replaced by TrimEnc₁. Compared with the TrimEnc defined in Figure 10, TrimEnc₁ uniformly samples ρ to encrypt m and no longer derive it by querying E.

As shown in Figure 10, when deriving the randomness ρ , the TrimEnc introduces a counter N and imposes that all the input to E end with ρ_0 . Therefore, if $\mathcal{A}$ has never queried E with inputs of the form $(*, * || \rho_0)$, the randomness derived by E in TrimEnc is uniformly random in $\mathcal{A}$'s view, and thus TrimEnc is equivalent to TrimEnc₁. At this point, since $\mathcal{A}$ queries E at most q_E times and ρ_0 is uniformly sampled from $\mathcal{R}$, by the One-Way to Hiding theorem [2] we have

$$|\Pr[1 \leftarrow \mathbf{G_0}] - \Pr[1 \leftarrow \mathbf{G_1}]| \leq \frac{2q_E}{\sqrt{|\mathcal{R}|}}. \quad (21)$$

Game $\mathbf{G_2}$: This game is the same as game $\mathbf{G_1}$, except that the key generation algorithm is replaced by TrimGen₁. Compared with the TrimGen defined in Figure 10, TrimGen₁ sets the repetition upper bound of the while loop as a positive integer s , i.e., if the (pk, sk) generated by the s -th repetition of KeyGen still results in $\text{Reject}_{key}(pk, sk) = 0$, game $\mathbf{G_2}$ aborts and outputs 0.

Obviously, as long as the event that all the instances of (pk, sk) generated by repeating KeyGen s times result in $\text{Reject}_{key}(pk, sk) = 0$ does not occur, game $\mathbf{G_1}$ and $\mathbf{G_2}$ are identical. Hence, by using the fundamental lemma [50] and the

definition of r_{key} given in Eq. (18), we have

$$|\Pr[1 \leftarrow \mathbf{G}_1] - \Pr[1 \leftarrow \mathbf{G}_2]| \leq (r_{\text{key}})^s. \quad (22)$$

Game $\mathbf{G}_3$: In this game, the TrimGen_1 is replaced by TrimGen_2 , which has the following two differences from TrimGen_1 :

1. TrimGen_2 no longer executes a while loop, but instead executes a for loop to generate $(pk_1, sk_1), \dots, (pk_s, sk_s)$.
2. Let Bad be the event that $(pk_1, sk_1), \dots, (pk_s, sk_s)$ satisfy $\text{Reject}_{\text{key}}(pk_1, sk_1) = \dots = \text{Reject}_{\text{key}}(pk_s, sk_s) = 0$. Then, if Bad occurs, game $\mathbf{G}_3$ aborts and outputs 0. Otherwise, there must exist $1 \leq i \leq s$ s.t. $\text{Reject}_{\text{key}}(pk_i, sk_i) = 1$ and $\text{Reject}_{\text{key}}(pk_j, sk_j) = 0$ for any $j < i$, and then TrimGen_2 outputs this (pk_i, sk_i) .

It is obvious that TrimGen_2 and TrimGen_1 are equivalent, as TrimGen_2 merely generates all the instances of (pk, sk) , which are generated sequentially via a while loop in TrimGen_1 , in one go. Hence, we have

$$\Pr[1 \leftarrow \mathbf{G}_2] = \Pr[1 \leftarrow \mathbf{G}_3]. \quad (23)$$

Game $\mathbf{G}_4$: In this game, the TrimGen_2 is replaced by TrimGen_3 , which is identical to TrimGen_2 except that if Bad does not occur, it takes (pk_l, sk_l) where $l \leftarrow_{\$} [s]$ as the final return.

By additionally writing out the $l \leftarrow_{\$} [s]$ used by game $\mathbf{G}_4$, we can compute

$$\begin{aligned} & \Pr[b = 1 : b \leftarrow \mathbf{G}_4, l \leftarrow_{\$} [s]] \\ & \stackrel{(a)}{=} \Pr[b = 1 \wedge \neg \text{Bad} : b \leftarrow \mathbf{G}_4, l \leftarrow_{\$} [s]] \\ & = \Pr[\neg \text{Bad}] \cdot \Pr[b = 1 : b \leftarrow \mathbf{G}_4, l \leftarrow_{\$} [s] | \neg \text{Bad}] \\ & \stackrel{(b)}{=} \Pr[\neg \text{Bad}] \cdot \frac{1}{s} \cdot \Pr[b = 1 : b \leftarrow \mathbf{G}_4, l = i | \neg \text{Bad}] \\ & \quad + \Pr[\neg \text{Bad}] \cdot \frac{s-1}{s} \cdot \Pr[b = 1 : b \leftarrow \mathbf{G}_4, l \neq i | \neg \text{Bad}] \\ & \stackrel{(c)}{=} \Pr[\neg \text{Bad}] \cdot \frac{1}{s} \cdot \Pr[b = 1 : b \leftarrow \mathbf{G}_3 | \neg \text{Bad}] \\ & \quad + \Pr[\neg \text{Bad}] \cdot \frac{s-1}{s} \cdot \Pr[b = 1 : b \leftarrow \mathbf{G}_4, l \neq i | \neg \text{Bad}] \\ & \geq \Pr[\neg \text{Bad}] \cdot \frac{1}{s} \cdot \Pr[b = 1 : b \leftarrow \mathbf{G}_3 | \neg \text{Bad}] \stackrel{(d)}{=} \frac{1}{s} \cdot \Pr[b = 1 : b \leftarrow \mathbf{G}_3]. \end{aligned}$$

Here (a) and (d) follow from the fact that outputting 1 by both game $\mathbf{G}_3$ and $\mathbf{G}_4$ implies that Bad does not occur. The parameter i introduces in (b) satisfies $\text{Reject}_{\text{key}}(pk_i, sk_i) = 1$ and $\text{Reject}_{\text{key}}(pk_j, sk_j) = 0$ for any $j < i$. Notice that such i must exist under the condition that Bad does not occur. (c) follows from the fact that, when $l = i$, the computation process of game $\mathbf{G}_3$ and $\mathbf{G}_4$ are actually identical according to their definitions. Now we have

$$\Pr[1 \leftarrow \mathbf{G}_3] \leq s \cdot \Pr[1 \leftarrow \mathbf{G}_4]. \quad (24)$$

Game $\mathbf{G}_5$: In this game, the TrimGen_3 is replaced by TrimGen_4 , which is identical to TrimGen_3 except that it no longer checks whether Bad occurs and directly returns (pk_l, sk_l) with $l \leftarrow_{\$} [s]$. Similar to the game hop $\mathbf{G}_1$ to $\mathbf{G}_2$, we have

$$|\Pr[1 \leftarrow \mathbf{G}_4] - \Pr[1 \leftarrow \mathbf{G}_5]| \leq \Pr[\text{Bad}] = (r_{\text{key}})^s. \quad (25)$$

Game $\mathbf{G}_6$: In this game, the TrimGen_4 is replaced by KeyGen , and the adversary $\mathcal{A}$ is replaced by the following adversary $\mathcal{B}$:

- $\mathcal{B}$ runs $\mathcal{A}$, simulates the random oracle $\mathbf{E}$ for $\mathcal{A}$ by using a $2q_{\mathbf{E}}$ -wise independent function, and finally returns $\mathcal{A}$'s output m .

It is not hard to see that game $\mathbf{G}_6$ is an equivalent rewrite of game $\mathbf{G}_5$. Thus

$$\Pr[1 \leftarrow \mathbf{G}_5] = \Pr[1 \leftarrow \mathbf{G}_6]. \quad (26)$$

Now by combining Eq. (20) to Eq. (26), we obtain

$$\text{Adv}_{\text{TrimPKE}, \mathcal{A}}^{\text{OW-CPA}} \leq s \cdot \Pr[1 \leftarrow \mathbf{G}_6] + (s+1) \cdot (r_{\text{key}})^s + \frac{2q_{\mathbf{E}}}{\sqrt{|\mathcal{R}|}}. \quad (27)$$

Next, by following a similar proof idea of the game sequence $\mathbf{G}_2$ to $\mathbf{G}_6$, we introduce the game sequence $\mathbf{G}_7$ to $\mathbf{G}_{11}$ as shown in Figure 15 to remove the rejection sampling of the TrimEnc_1 used by game $\mathbf{G}_6$.

GAMES $\mathbf{G}_7$ - $\mathbf{G}_{11}$	$\text{TrimEnc}_3(pk, m)$ // $\mathbf{G}_8$	$\text{TrimEnc}_4(pk, m)$ // $\mathbf{G}_9$
1 : $(pk, sk) \leftarrow \text{KeyGen}$ // $\mathbf{G}_7$ - $\mathbf{G}_{11}$	1 : for k from 1 to t	1 : for k from 1 to t
2 : $m^* \leftarrow_{\$} \mathcal{M}$ // $\mathbf{G}_7$ - $\mathbf{G}_{11}$	$\rho_k \leftarrow_{\$} \mathcal{R}$	$\rho_k \leftarrow_{\$} \mathcal{R}$
$c^* \leftarrow \text{TrimEnc}_{i-5}(pk, m^*)$ // $\mathbf{G}_{7 \leq i \leq 10}$	$c_k \leftarrow \text{Encrypt}(pk, m, \rho_k)$	$c_k \leftarrow \text{Encrypt}(pk, m, \rho_k)$
$c^* \leftarrow \text{Encrypt}(pk, m^*)$ // $\mathbf{G}_{11}$	end for	end for
3 : $m \leftarrow \mathcal{B}(pk, c^*)$ // $\mathbf{G}_7$ - $\mathbf{G}_{11}$	2 : if Bad' occurs, abort	2 : if Bad' occurs, abort
4 : return $[m = m^*]$ // $\mathbf{G}_7$ - $\mathbf{G}_{11}$	game return 0	game return 0
	else return c_i	else
$\text{TrimEnc}_2(pk, m)$ // $\mathbf{G}_7$		$l \leftarrow_{\$} [t]$
1 : $n = 1, \rho \leftarrow_{\$} \mathcal{R}$	$\text{TrimEnc}_5(pk, m)$ // $\mathbf{G}_{10}$	return c_l
$c \leftarrow \text{Encrypt}(pk, m, \rho)$	1 : for k from 1 to t	
2 : while $\text{Reject}_{\text{enc}}(pk, m, \rho, c) = 0 \wedge n < t$	$\rho_k \leftarrow_{\$} \mathcal{R}$	
$n = n + 1$	$c_k \leftarrow \text{Encrypt}(pk, m, \rho_k)$	
$\rho \leftarrow_{\$} \mathcal{R}$	end for	
$c \leftarrow \text{Encrypt}(pk, m, \rho)$	2 : $l \leftarrow_{\$} [t]$	
end while	return c_l	
3 : if $\text{Reject}_{\text{enc}}(pk, m, \rho, c) = 0$, abort		
game return 0		
else return c		

Fig. 15: Games $\mathbf{G}_7$ to $\mathbf{G}_{11}$ in the proof of Theorem 2.

Game $\mathbf{G_7}$: This game is the same as game $\mathbf{G_6}$, except that the encryption algorithm TrimEnc_1 is replaced by TrimEnc_2 . Compared with the TrimEnc_1 defined in Figure 10, TrimEnc_2 sets the repetition upper bound of the while loop as a positive integer t , i.e., if the (ρ, c) generated by the t -th repetition still results in $\text{Reject}_{\text{enc}}(pk, m, \rho, c) = 0$, game $\mathbf{G_7}$ aborts and outputs 0.

Similar to the proof of the game hop $\mathbf{G_1}$ to $\mathbf{G_2}$, by the definition of r_{enc} given in Eq. (18), we have

$$|\Pr[1 \leftarrow \mathbf{G_6}] - \Pr[1 \leftarrow \mathbf{G_7}]| \leq (r_{\text{enc}})^t. \quad (28)$$

Game $\mathbf{G_8}$: In this game, the TrimEnc_2 is replaced by TrimEnc_3 , which has the following two differences from TrimEnc_2 :

1. TrimEnc_3 no longer executes a while loop, but instead executes a for loop to generate $(\rho_1, c_1), \dots, (\rho_t, c_t)$.
2. Let Bad' be the event that $(\rho_1, c_1), \dots, (\rho_t, c_t)$ satisfy $\text{Reject}_{\text{enc}}(pk, m, \rho_1, c_1) = \dots = \text{Reject}_{\text{enc}}(pk, m, \rho_t, c_t) = 0$. Then, if Bad' occurs, game $\mathbf{G_8}$ aborts and outputs 0. Otherwise, there must exist $1 \leq i \leq t$ s.t. $\text{Reject}_{\text{enc}}(pk, m, \rho_i, c_i) = 1$ and $\text{Reject}_{\text{enc}}(pk, m, \rho_j, c_j) = 0$ for any $j < i$, and then TrimEnc_3 takes this c_i as the final return.

Obviously, game $\mathbf{G_8}$ is an equivalent rewrite of game $\mathbf{G_7}$, and thus

$$\Pr[1 \leftarrow \mathbf{G_7}] = \Pr[1 \leftarrow \mathbf{G_8}]. \quad (29)$$

Game $\mathbf{G_9}$: In this game, the TrimEnc_3 is replaced by TrimEnc_4 , which is identical to TrimEnc_3 except that if Bad' does not occur, it takes c_l where $l \leftarrow_{\$} [t]$ as the final return. Similar to the proof of the game hop $\mathbf{G_3}$ to $\mathbf{G_4}$, we can compute

$$\Pr[1 \leftarrow \mathbf{G_8}] \leq t \cdot \Pr[1 \leftarrow \mathbf{G_9}]. \quad (30)$$

Game $\mathbf{G_{10}}$: In this game, the TrimEnc_4 is replaced by TrimEnc_5 , which is identical to TrimEnc_4 except that it no longer checks whether Bad' occurs and directly returns c_l with $l \leftarrow_{\$} [t]$. Similar to the game hop $\mathbf{G_6}$ to $\mathbf{G_7}$, we have

$$|\Pr[1 \leftarrow \mathbf{G_9}] - \Pr[1 \leftarrow \mathbf{G_{10}}]| \leq \Pr[\text{Bad}'] = (r_{\text{enc}})^t. \quad (31)$$

Game $\mathbf{G_{11}}$: In this game, the TrimEnc_5 is replaced by Encrypt . Obviously we have $\mathcal{T}_{\mathcal{B}} \approx \mathcal{T}_{\mathcal{A}}$, and game $\mathbf{G_{11}}$ is an equivalent rewrite of game $\mathbf{G_{10}}$. In addition, game $\mathbf{G_{11}}$ can be also viewed as an OW-CPA game of PKE with the adversary $\mathcal{B}$, and thus

$$\Pr[1 \leftarrow \mathbf{G_{10}}] = \Pr[1 \leftarrow \mathbf{G_{11}}] = \text{Adv}_{\text{PKE}, \mathcal{B}}^{\text{OW-CPA}}. \quad (32)$$

Now by combining Eq. (27) to Eq. (32), we can obtain

$$\text{Adv}_{\text{TrimPKE}, \mathcal{A}}^{\text{OW-CPA}} \leq st \cdot \text{Adv}_{\text{PKE}, \mathcal{B}}^{\text{OW-CPA}} + (st + s) \cdot (r_{\text{enc}})^t + (s + 1) \cdot (r_{\text{key}})^s + \frac{2q_{\text{E}}}{\sqrt{|\mathcal{R}|}}.$$

This completes the proof of Theorem 2. $\square$

6.2 IND-CCA Security Proof of TrimKEM in the QROM

Notably, the KEM scheme TrimKEM shown in Figure 13 is obtained by applying the FO transformation to the public-key encryption scheme TrimPKE. Hence, by combining the security proofs of FO given in [31,35,47] with the above Theorem 2, we have the following security theorems for TrimKEM.

Theorem 3. *Let $\mathcal{A}$ be an IND-CCA adversary against TrimKEM, making q_D classical queries to the decapsulation oracle, and making q_E, q_F, q_G and q_H quantum queries to the random oracle E, F, G and H, respectively. Then we can construct an OW-CPA adversary $\mathcal{B}$ against PKE such that $\mathcal{T}_{\mathcal{B}} \approx \mathcal{T}_{\mathcal{A}}$ and*

$$\begin{aligned} \text{Adv}_{\text{TrimKEM}, \mathcal{A}}^{\text{IND-CCA}} \leq & 2(q_G + q_F) \sqrt{st \cdot \text{Adv}_{\text{PKE}, \mathcal{B}}^{\text{OW-CPA}} + (st + s) \cdot (r_{\text{enc}})^t + (s + 1) \cdot (r_{\text{key}})^s + \frac{2q_E}{\sqrt{|\mathcal{R}|}}} \\ & + \frac{11q_F + 2q_G + 2q_H}{\sqrt{|\mathcal{M}|}} + 4q_G \cdot \sqrt{\delta_{\text{TrimPKE}}} \end{aligned}$$

holds for any positive integers s and t . Here r_{key} and r_{enc} are defined in Eq. (18), $\mathcal{M}/\mathcal{R}$ is the message/randomness space of PKE, and δ_{TrimPKE} is the worst-case DFR of TrimPKE.

Theorem 3 demonstrates that, after applying our Trim-protection to PKE to obtain TrimPKE, we can indeed obtain an IND-CCA-secure KEM scheme in the QROM via the FO transformation based on the OW-CPA security of PKE and the worst-case DFR of TrimPKE. Notice that the security bound of Theorem 3 holds for any positive integers s and t . Therefore, as long as the rejection probabilities r_{key} and r_{enc} are relatively small, e.g., a fixed constant of 1%, the rejection sampling introduced by trim will have no essential impact on the security.

We point out that the first square-root of the security bound of Theorem 3 can be avoided if we instead combine the tighter IND-CCA security proofs of FO given in [38,26]. In addition, if we apply the transformation $\text{FOAC}_0 (= \text{FO} \circ \text{ACWC}_0)$ defined in [18,27] to TrimPKE to obtain the TrimKEM, the worst-case DFR of TrimPKE (i.e. δ_{TrimPKE}) used in the security bound of Theorem 3 could be replaced by the average-case DFR of TrimPKE.

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A Algorithms

A.1 TrimLAC

We keep the notation of the trim upper bound as B_{inner} . Let $\Psi_{n,hw}$ denote the distribution of the polynomial whose coefficients contain $hw/2$ 1s and $hw/2$ -1s, and the rest are all 0s. For a polynomial a , we use $\lfloor a \rfloor$ to denote the polynomial after the rounding operation such that $\lfloor a \rfloor_{[i]} = \lfloor a_{[i]} \rfloor$. The functions ECCenc and ECCdec are the encoding and decoding functions corresponding to the ECC. We use the symmetric primitives as: a pseudorandom function PRF, an extendable-output function XOF, and two variable-length input hash functions G and H. In Algorithms 2 and 3, we do not replicate the full framework of TrimPKE 10; instead, we only adapt the relevant parts. Since the seeds ρ_a and (ρ_s, ρ_e) are decoupled, and rejection sampling does not apply to the sampling of a , the concrete algorithms can derive randomness partially while still following the same deterministic-reseeding pattern. We use these appendix algorithms only as case-study instantiations of the generic framework; the load-bearing reduction remains the generic TrimPKE/TrimKEM proof.

Algorithm 2 TrimLAC.PKE.KeyGen

Output: the public key pk , the private key sk

- 1: $\rho_a \leftarrow \text{Sample}(\mathcal{U}(\{0, 1\}^{256}))$
 - 2: $a \leftarrow \text{Sample}(\mathcal{U}(\mathcal{R}_n/q\mathcal{R}_n), \rho_a)$
 - 3: $\rho_s \leftarrow \text{Sample}(\mathcal{U}(\{0, 1\}^{256}))$
 - 4: $s \leftarrow \text{Sample}(\Psi_{n,hw})$
 - 5: **if** $\|(s\bar{s})_{[1:n/2]}\|_\infty \geq B_{\text{inner}}$
 - 6: go back to step 4
 - 7: $\rho_e \leftarrow \text{Sample}(\mathcal{U}(\{0, 1\}^{256}))$
 - 8: $e \leftarrow \text{Sample}(\Psi_{n,hw})$
 - 9: **if** $\|(e\bar{e})_{[1:n/2]}\|_\infty \geq B_{\text{inner}}$
 - 10: go back to step 8
 - 11: $b \leftarrow as + e \pmod{q}$
 - 12: **return** $(pk := (\rho_a, b), sk := s)$
-

Algorithm 3 TrimLAC.PKE.Encrypt

Input: the public key $pk := (\rho_a, b)$, the message m , the seed ρ_0

Output: the ciphertext c

```
1:  $a \leftarrow \text{Sample}(\mathcal{U}(\mathcal{R}_n/q\mathcal{R}_n), \rho_a)$ 
2:  $\tilde{m} \leftarrow \text{ECCEnc}(m)$ 
3:  $\rho \leftarrow \rho_0$ 
4:  $N_\rho \leftarrow 0$ 
5: repeat
6:    $\rho \leftarrow \text{H}(N_\rho \parallel \rho \parallel \rho_0)$ 
7:    $N_\rho \leftarrow N_\rho + 1$ 
8:    $N_{\text{PRF}} \leftarrow 0$ 
9:    $r \leftarrow \text{Sample}(\Psi_{n,hw}, \text{PRF}(\rho, N_{\text{PRF}}))$ 
10:   $N_{\text{PRF}} \leftarrow N_{\text{PRF}} + 1$ 
11:   $e_1 \leftarrow \text{Sample}(\Psi_{n,hw}, \text{PRF}(\rho, N_{\text{PRF}}))$ 
12:   $N_{\text{PRF}} \leftarrow N_{\text{PRF}} + 1$ 
13:   $e_2 \leftarrow \text{Sample}(\Psi_{n,hw}, \text{PRF}(\rho, N_{\text{PRF}}))$ 
14: until  $\|(r\tilde{r})_{[1:n/2]}\|_\infty < B_{\text{inner}}$  and  $\|(e_1\tilde{e}_1)_{[1:n/2]}\|_\infty < B_{\text{inner}}$ 
15:  $c_1 \leftarrow ar + e_1 \pmod{q}$ 
16:  $c_2 \leftarrow br + e_2 + \lfloor q \rfloor \cdot \tilde{m} \pmod{q}$ 
17: return  $c := (c_1, c_2)$ 
```

Algorithm 4 TrimLAC.PKE.Decrypt

Input: the private key $sk := s$, the ciphertext $c := (c_1, c_2)$

Output: the message m

```
1:  $\tilde{m} \leftarrow \lfloor \frac{2}{q} \cdot (c_2 - c_1 s \pmod{q}) \rfloor \pmod{2}$ 
2:  $m \leftarrow \text{ECCDec}(\tilde{m})$ 
3: return  $m$ 
```

Algorithm 5 TrimLAC.KEM.KeyGen

Output: the public key pk , the private key sk

```
1:  $(pk, sk') \leftarrow \text{TrimLAC.PKE.KeyGen}$ 
2:  $\bar{K}' \leftarrow \text{Sample}(\mathbb{Z}_2^{256})$ 
3:  $sk \leftarrow (sk', pk, \text{H}(pk), \bar{K}')$ 
4: return  $(pk, sk)$ 
```

Algorithm 6 TrimLAC.KEM.Encapsulate

Input: the public key pk

Output: the ciphertext c , the shared key K

```
1:  $m \leftarrow \text{Sample}(\mathbb{Z}_2^{256})$ 
2:  $(\bar{K}, \rho_0) \leftarrow \text{G}(m \parallel \text{H}(pk))$ 
3:  $c \leftarrow \text{TrimLAC.PKE.Encrypt}(pk, m, \rho_0)$ 
4:  $K \leftarrow \text{KDF}(\bar{K} \parallel \text{H}(c))$ 
5: return  $c, K$ 
```

Algorithm 7 TrimLAC.KEM.Decapsulate

Input: the private key sk , the ciphertext c

Output: the shared key K

```
1:  $(sk', pk, H(pk), \bar{K}') \leftarrow sk$ 
2:  $m' \leftarrow \text{TrimLAC.PKE.Decrypt}(sk', c)$ 
3:  $(\bar{K}, \rho_0) \leftarrow G(m' \| H(pk))$ 
4: if  $c \neq \text{TrimLAC.PKE.Encrypt}(pk, m', \rho_0)$ 
5:    $K \leftarrow \text{KDF}(\bar{K}' \| H(c))$ 
6: else
7:    $K \leftarrow \text{KDF}(\bar{K} \| H(c))$ 
8: return  $K$ 
```

A.2 TrimDAWN

We set the trim parameters as $\omega_f, \omega_g, \omega_s, \omega_e, \omega_{e_{dc}}$. Let $\theta_f, \theta_g, \theta_s, \theta_e, \theta_{e_{dc}}$ denote the corresponding calibration tuples used by `IsDistorted`. For reference, the underlying coefficient samplers have standard deviations $\sigma_f, \sigma_g, \sigma_s, \sigma_e, \sigma_{e_{dc}}$ for the distributions $\mathcal{S}_{n,k_g}, \mathcal{S}_{n,k_f}, \mathcal{S}_{n,k_s}, \mathcal{S}_{n,k_e}, \mathcal{X}_{e_{dc}}$, where $\mathcal{S}_{n,k}$ corresponds to the sparse ternary distribution $\{0 : 1 - \frac{2k}{n}, 1 : \frac{k}{n}, -1 : \frac{k}{n}\}$ and $\mathcal{X}_{e_{dc}}$ corresponds to the distribution of the noise polynomial e_{dc} introduced by ciphertext compression. We use $(\text{mod } \langle \varphi, \mathbb{Z}_q \rangle)$ to denote modular arithmetic in the quotient ring $\mathbb{Z}_q[x]/\varphi$. Throughout this subsection, we follow the notation of DAWN [42] whenever possible for consistency.

Algorithm 8 SimpleDecoding

Input: three polynomials $\mathbf{c}', \mathbf{m}', \mathbf{f}_2 \in \mathcal{R}_{x^{n/2}+1}$

Output: the message polynomial $\mathbf{m}$

```

1:  $\mathbf{e}' \leftarrow (\mathbf{c}') \pmod{\langle x^{n/4} + 1, 2 \rangle}$ 
2:  $\mathbf{c}'' \leftarrow ((\mathbf{e}')^{-1} \mathbf{c}') \pmod{\langle x^n + 1 \rangle}$ 
3:  $i \leftarrow 0$ 
4:  $base \leftarrow q$ 
5: for  $j$  from 0 to 3
6:    $v \leftarrow \min(|\mathbf{c}''_{[jn/4]} + q|, |\mathbf{c}''_{[jn/4]} - q|)$ 
7:   if  $v < base$ 
8:      $base \leftarrow v$ 
9:      $i \leftarrow jn/4$ 
10:  $\mathbf{m} \leftarrow (\mathbf{m}' + x^i \mathbf{f}_2 \mathbf{e}') \pmod{\langle x^{n/2} + 1, 2 \rangle}$ 
11: if  $\mathbf{e}' = 0$ 
12:    $\mathbf{m} \leftarrow (\mathbf{m}') \pmod{\langle x^{n/4} + 1, 2 \rangle}$ 
13: else
14:    $\mathbf{m} \leftarrow (\mathbf{m}) \pmod{\langle x^{n/4} + 1, 2 \rangle}$ 
15: return  $\mathbf{m}$ 

```

Algorithm 9 TrimDAWN.PKE.KeyGen

Output: the public key pk , the private key sk

- 1: $g \leftarrow \text{Sample}(\mathcal{S}_{n,k_g}^n)$
 - 2: $f \leftarrow \text{Sample}(\mathcal{S}_{n,k_f}^n)$
 - 3: **if** f or g is not invertible in $\mathcal{R}_n/q\mathcal{R}_n$
 - 4: restart key generation
 - 5: **if** $\text{IsDistorted}(g, \theta_g, \omega_g) + \text{IsDistorted}(f, \theta_f, \omega_f) \neq 0$
 - 6: restart key generation
 - 7: $f_2 \leftarrow f^{-1} \pmod{\langle x^{n/2} + 1, 2 \rangle}$
 - 8: **if** f_2 is not invertible in $\mathcal{R}_{n/2}/2\mathcal{R}_{n/2}$
 - 9: restart key generation
 - 10: $h \leftarrow (gf^{-1}) \pmod{\langle x^n + 1, q \rangle}$
 - 11: $pk \leftarrow h$
 - 12: $sk \leftarrow (f, f_2)$
 - 13: **return** (pk, sk)
-

Algorithm 10 TrimDAWN.PKE.Encrypt

Input: the public key pk , the message m , the seed ρ_0

Output: the ciphertext c

- 1: $h \leftarrow pk$
 - 2: $\rho \leftarrow \rho_0$
 - 3: $N_\rho \leftarrow 0$
 - 4: **repeat**
 - 5: $\rho \leftarrow \text{H}(N_\rho \parallel \rho \parallel \rho_0)$
 - 6: $N_\rho \leftarrow N_\rho + 1$
 - 7: $N_{\text{PRF}} \leftarrow 0$
 - 8: $s \leftarrow \text{Sample}(\mathcal{S}_{n,k_s}^n, \text{PRF}(\rho, N_{\text{PRF}}))$
 - 9: $N_{\text{PRF}} \leftarrow N_{\text{PRF}} + 1$
 - 10: $e \leftarrow \text{Sample}(\mathcal{S}_{n,k_e}^n, \text{PRF}(\rho, N_{\text{PRF}}))$
 - 11: $N_{\text{PRF}} \leftarrow N_{\text{PRF}} + 1$
 - 12: $c' \leftarrow \left(hs + e + (x^{n/4} + 1)^{-1} m \right) \pmod{\langle x^n + 1, q \rangle}$
 - 13: $c \leftarrow \lfloor \frac{1}{d_c} \cdot c' \rfloor$
 - 14: $e_{d_c} \leftarrow c' - d_c \cdot c$
 - 15: **until** $\text{IsDistorted}(s, \theta_s, \omega_s) = \text{IsDistorted}(e, \theta_e, \omega_e) = \text{IsDistorted}(e_{d_c}, \theta_{e_{d_c}}, \omega_{e_{d_c}}) = 0$
 - 16: **return** c
-

Algorithm 11 TrimDAWN.PKE.Decrypt

Input: the ciphertext c , the private key sk

Output: the message m

- 1: $m \leftarrow \text{DAWN.PKE.Decrypt}(c, sk)$
 - 2: **return** m
-

Algorithm 12 TrimDAWN.KEM.KeyGen

Output: the public key pk , the private key sk

- 1: $(pk, sk') \leftarrow \text{TrimDAWN.PKE.KeyGen}$
 - 2: $K' \leftarrow \text{Sample}(\mathcal{U}(\mathcal{M}))$
 - 3: $sk \leftarrow (K', sk', pk, H(pk))$
 - 4: **return** (pk, sk)
-

Algorithm 13 TrimDAWN.KEM.Encapsulation

Input: the public key pk

Output: the ciphertext c , the shared key K

- 1: $m \leftarrow \text{Sample}(\mathcal{M})$
 - 2: $(\bar{K}, \rho_0) \leftarrow G(m \| H(pk))$
 - 3: $c \leftarrow \text{TrimDAWN.PKE.Encrypt}(pk, m, \rho_0)$
 - 4: $K \leftarrow \text{KDF}(\bar{K}, H(c))$
 - 5: **return** (c, K)
-

Algorithm 14 TrimDAWN.KEM.Decapsulation

Input: the ciphertext c , the private key sk

Output: the shared key K

- 1: $(K', sk', pk, H(pk)) \leftarrow sk$
 - 2: $m \leftarrow \text{TrimDAWN.PKE.Decrypt}(c, sk')$
 - 3: $(\bar{K}, \rho_0) \leftarrow G(m \| H(pk))$
 - 4: **if** $c \neq \text{TrimDAWN.PKE.Encrypt}(pk, m, \rho_0)$
 - 5: $K \leftarrow \text{KDF}(\bar{K}', H(c))$
 - 6: **else**
 - 7: $K \leftarrow \text{KDF}(\bar{K}, H(c))$
 - 8: **return** K
-

B Security Proof